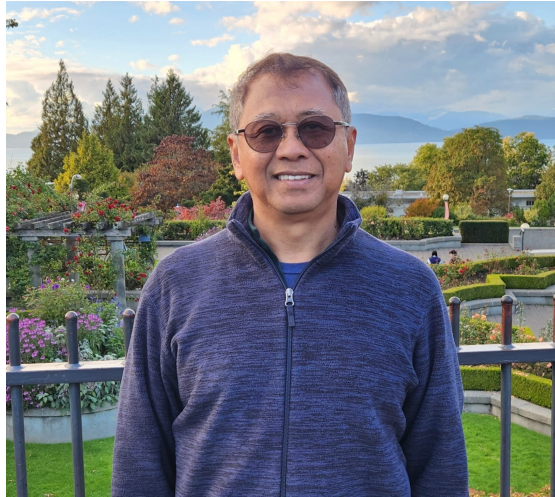




PROFESSIONAL SUMMARY

Senior Drilling Engineer with more than 32 years of extensive experience in the Oil & Gas industry, including 29 years with Operators and 3 years with Drilling Contractors, supported by a Mechanical Engineering background from Sepuluh Nopember Institute of Technology (ITS), Surabaya, Indonesia.

Possesses broad technical and operational expertise across the full drilling lifecycle with proven experience in Deepwater Exploration & Development Drilling, Shelf (Shallow Water) Development Drilling, and Onshore Exploration Projects. Demonstrated capability in leading and delivering complex drilling programs through integrated planning, execution management, technical assurance, and operational supervision.

Extensive experience serving as an independent and competent drilling professional with strong exposure to multidisciplinary coordination across Subsurface, Wells, Facilities, and Operations disciplines to optimize technical and commercial outcomes.

Demonstrated successful performance as Drilling Superintendent for challenging onshore gas exploration campaigns utilizing HPWBM mud systems in fractured basement geological environments, including execution of well testing activities resulting in successful gas discovery. Recognized for strong technical capability, analytical problem solving, operational leadership, and the ability to challenge conventional approaches to deliver practical, safe, and economically optimized solutions.

Proficient in engineering and drilling software including Landmark StressCheck, Landmark WellPlan Torque & Drag, and WellView, supported by strong competency in Microsoft Office Suite (Excel, Word, and PowerPoint). Demonstrates excellent communication capability in English (oral and written) and effective collaboration within multicultural and multidisciplinary project environments.

Torque & Drag

Objectives

At the end of this module you should:

- Understand the theories Torque & Drag Modeling
- Appreciate the importance of Side Force and its effect on Friction Forces
- Understand the role of Friction Factors and Tortuosity in modeling
- Be able to run Single point and multi point simulations in Drilling Office and interpret the outputs
- Build a Broomstick Plot for monitoring trends
- Understand the mechanisms to reduce Torque and Drag

Torque and Drag Uses (why do the modeling?)

- Define rig requirements
- Determine drillability of the well
- Optimize the trajectory and BHA / drill string design
- Identify problem areas
- Establish mud program needs
- Evaluate the effectiveness of hole cleaning actions
- Determining reaming, backreaming and short trip requirements
- Can simulate drilling and also completion (casing) runs

Torque and Drag Modeling

To understand computer modeling following the must be understood:

Model (Representation) – noun(C):

a representation of something, either as a physical object which is usually smaller than the real object, or as a **simple description** of the object which might be used in calculations.

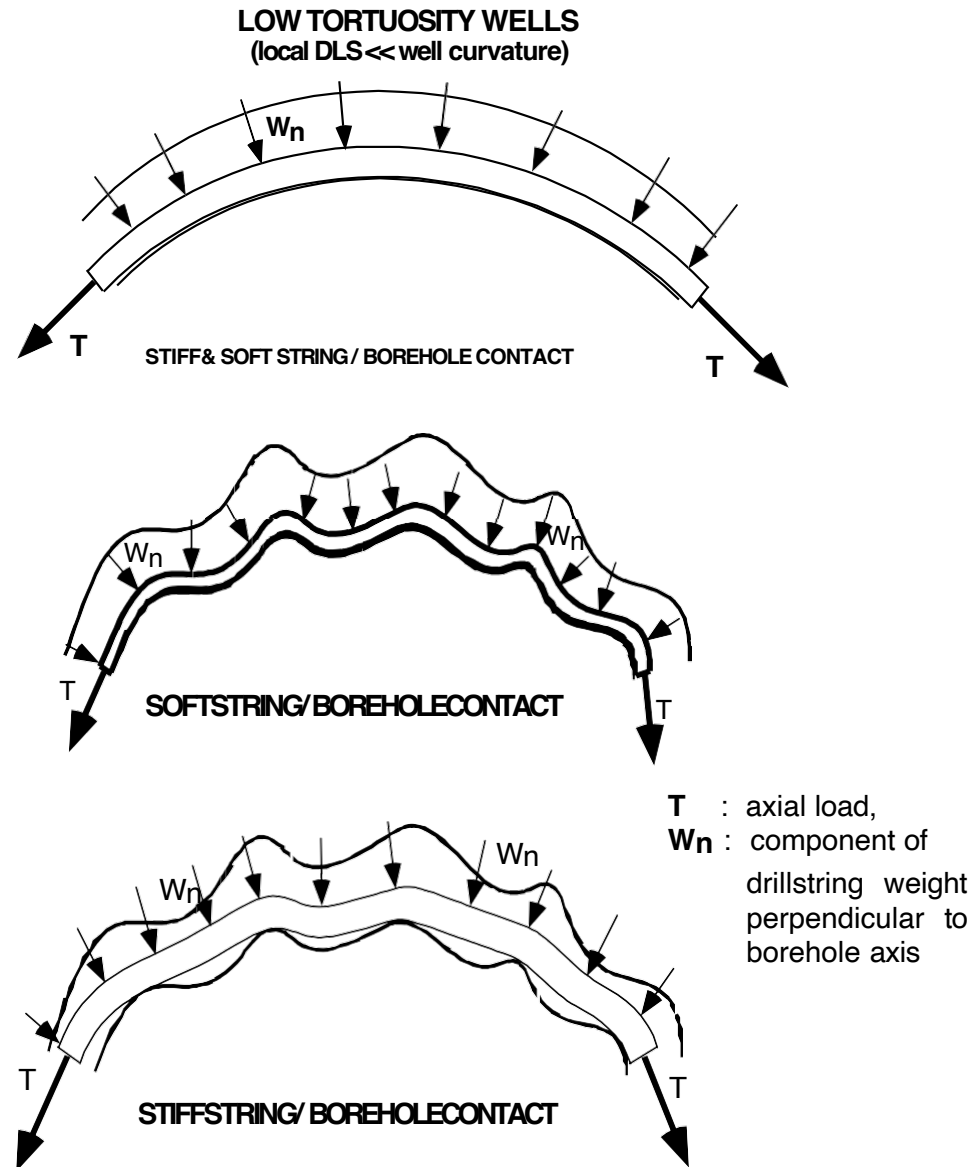
Garbage In = Garbage Out

Stiff vs. Soft String Model

Soft String	Stiff String
■ Drill string always in contact with the borehole ■ Contact area, curvature side forces are overestimated	■ Drill string curvature can be different than wellbore ■ Contact areas are reduced, more realistic side forces ■ More accurate torque loss calculation in a low inclination wellbore

Borehole/Drill string contact

- Stiff- and soft-string models give the same result for an untortuous plan



Torque and Drag Modeling- Inputs

1. Wellbore Geometry
2. Trajectory
3. BHA and drill stem details
4. Friction Factors
5. Mud properties
6. Tortuosity (to account for actual being worse than planned)
7. Scenario details (drilling, reaming, tripping)
8. Whether single point or multi point analysis

T & D Simulation

- **Single Point analysis**

- Bit at specified depth, output -> Axial Load, Side forces, Torque, Displacement and V.M stresses throughout entire drill string from bit to surface.

- **Multi-Point Analysis**

- Bit depth stepped down at intervals, output -> surface values of Hook Load and Torque i.e broomstick plots.

Something Additional!!

Tortuosity in Planned Trajectories

Why add tortuosity to plans?

- Account for more than “Ideal” T&D numbers
- Allows more consistent results between different engineers
- Account for drilling system used

Recommended Values (no offset data)

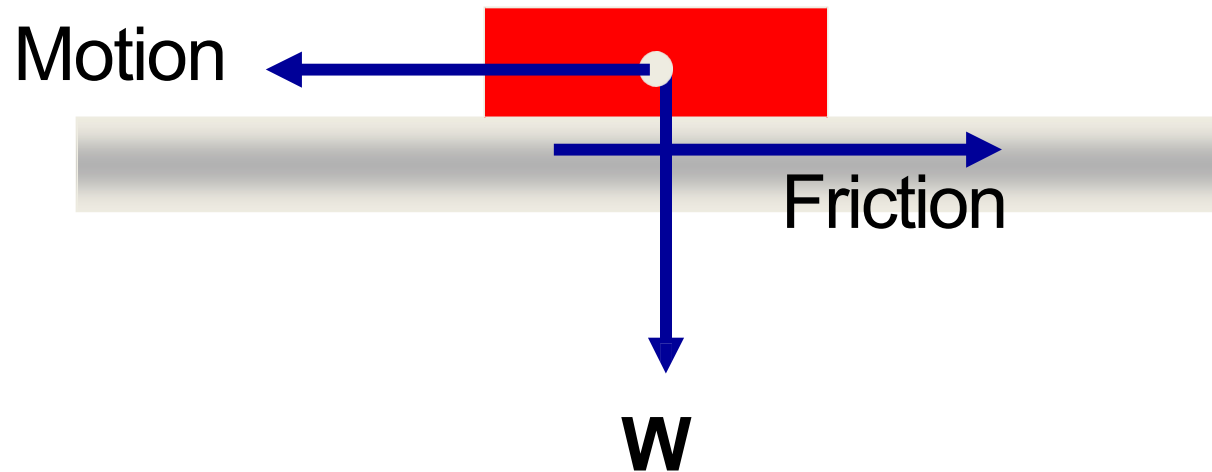
- Vertical, tangent sections 0.75/100ft period
- Build, drop sections 1.5/100ft period
- Turn only sections 1.0/100ft period

Components Of Torque & Drag

Side Force's & Friction

Friction

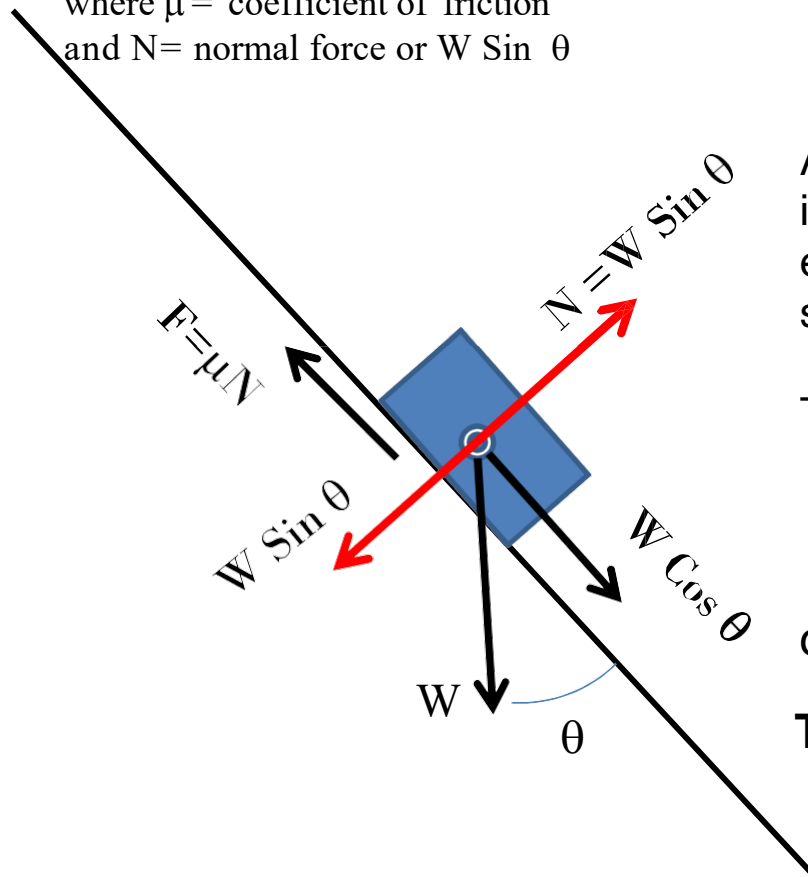
It is the resistance to motion that exists when an object is moved tangentially with respect to another which it touches.



Coefficient of Friction and critical Inclination

Friction Force, $F = \mu N$

where μ = coefficient of friction
and N = normal force or $W \cos \theta$



As θ increases there comes a critical inclination where the friction force, F exactly equals the component of weight down the slope

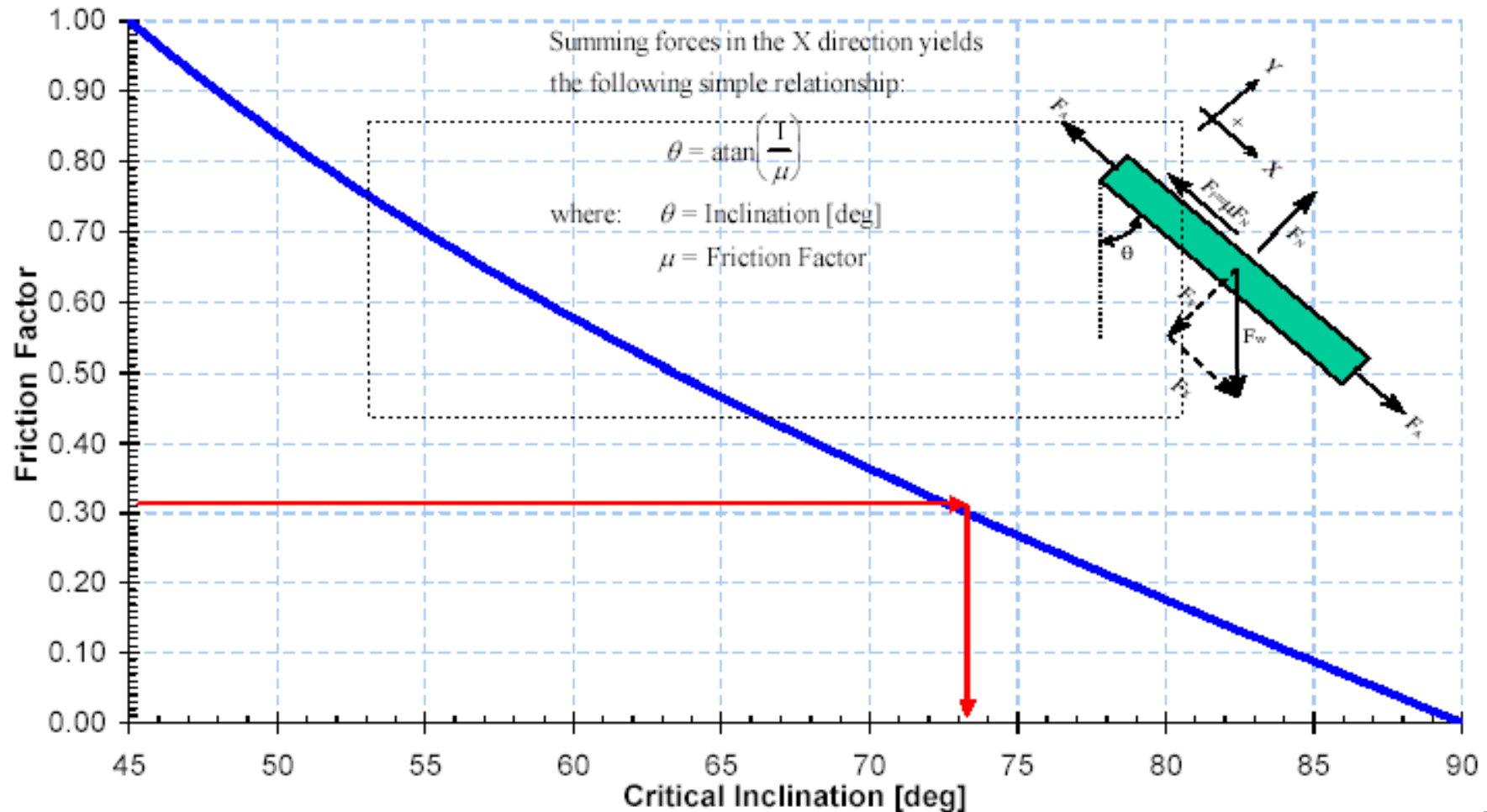
Thus at critical θ :

$$F = \mu W \sin \theta = W \cos \theta$$

$$\text{or } \mu = \cos \theta / \sin \theta = 1 / \tan \theta$$

$$\text{Thus critical Inclination} = \tan^{-1} (1/\mu)$$

When does the Pipe Stop Moving under own weight?



$$\tan^{-1}(1/\mu) = \text{Critical Inclination}$$

Exercise 1

What is the maximum hole angle (inclination angle) that can be logged without the aid of drillpipe, coiled tubing, other tubulars or sinker bars?
(assume $FF = 0.4$)

Friction Factors

In T & D modeling we use Friction Factor instead of coefficient of friction, μ

Friction Factor (FF) is really a Fudge Factor used in modeling and is not a true sliding coefficient of friction. It takes into account a wide variety of factors too complicated or not cost effective to model. These include:

- Formation friction
- Mud type (lubricity)
- Wellbore Tortuosity
- Borehole Condition (spiraled, ledged)
- Polishing of inside of casing etc.

Friction Factors

Notes on friction factors in Drilling Office:

Drilling Office uses dynamic FF as opposed to initial static FF which is higher

FF is independent of velocity but to get an object moving an initial higher static friction has to be overcome

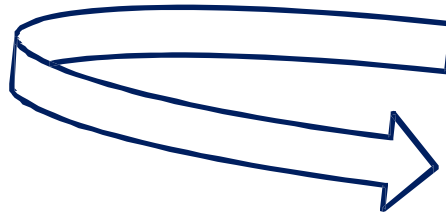
Friction Factors

Two different categories of friction factor defined in drilling office

- A. Translational Friction Factor** – dynamic FF opposing the motion into or out of the wellbore (used when tripping or sliding)

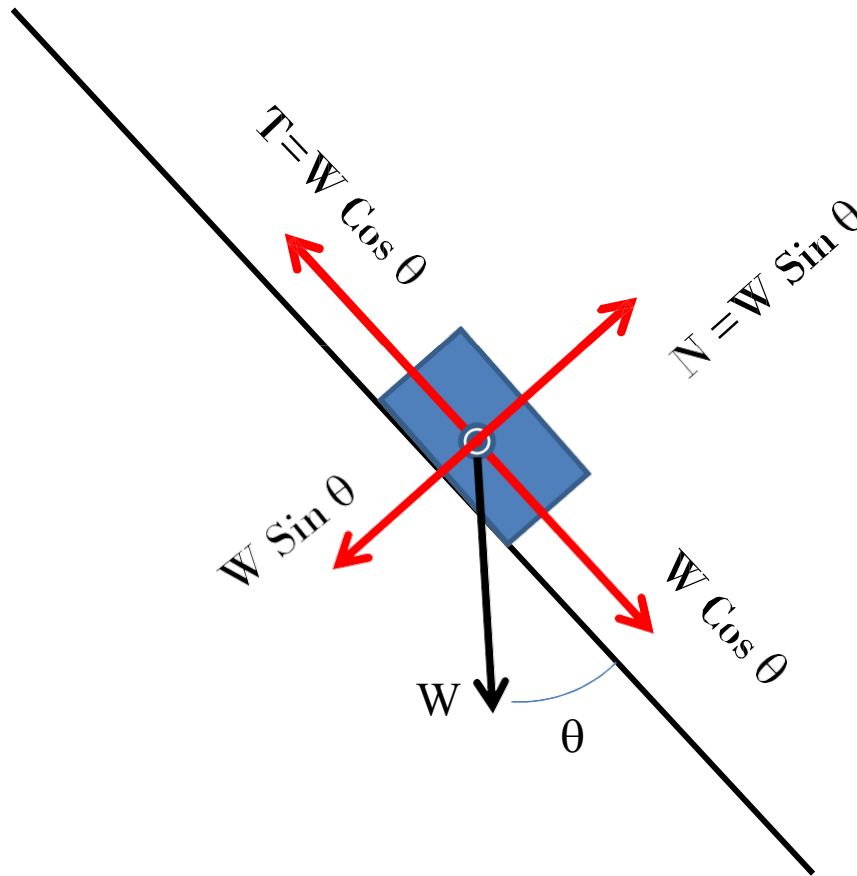


- B. Rotational Friction Factor** – dynamic FF opposing the rotation within the wellbore (used when drilling)



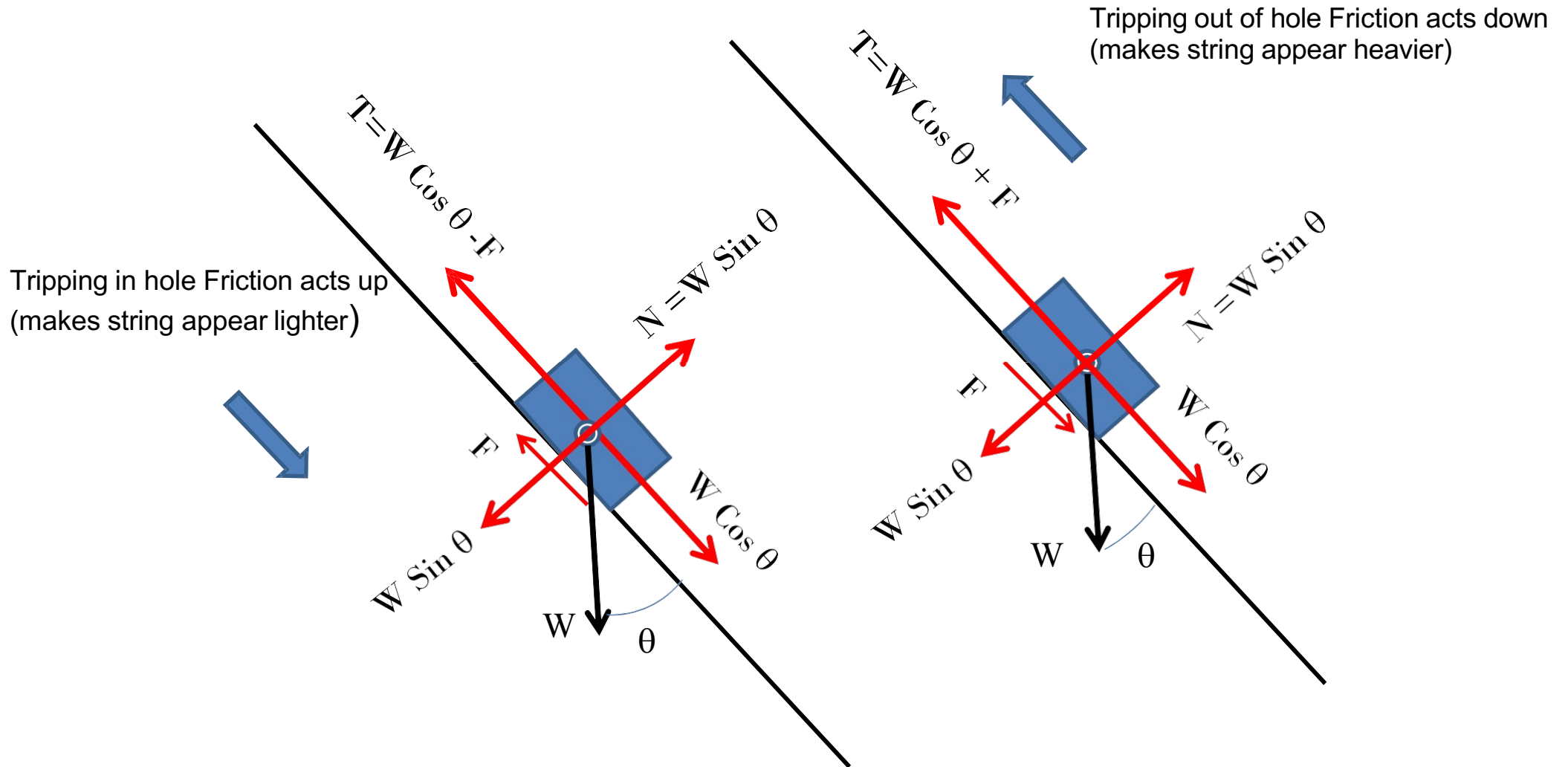
These values are inputted separately into drilling office (based on offset historical data) and the software computes the resultant value based on RPM and ROP inputs.

Effect of Friction (no doglegs)



No motion up nor down but
pipe rotating (friction = 0)

Effect of Friction (no doglegs)



Effect of Friction (no doglegs)

(a) Lowering: Friction opposes motion, so

$$\begin{aligned} T &= W \cos \theta - F \\ &= W \cos \theta - FF \times W \sin \theta \end{aligned}$$

F = Friction Force
 FF = Friction Factor

(b) Raising: Friction still opposes motion

$$\begin{aligned} T &= W \cos \theta + F \\ &= W \cos \theta + FF \times W \sin \theta \end{aligned}$$

This explains difference between Up weight, Down weight and Neutral (Rotating) weight

Components of Side Force

Weight always a consideration, use light drill pipe in Horizontal wells

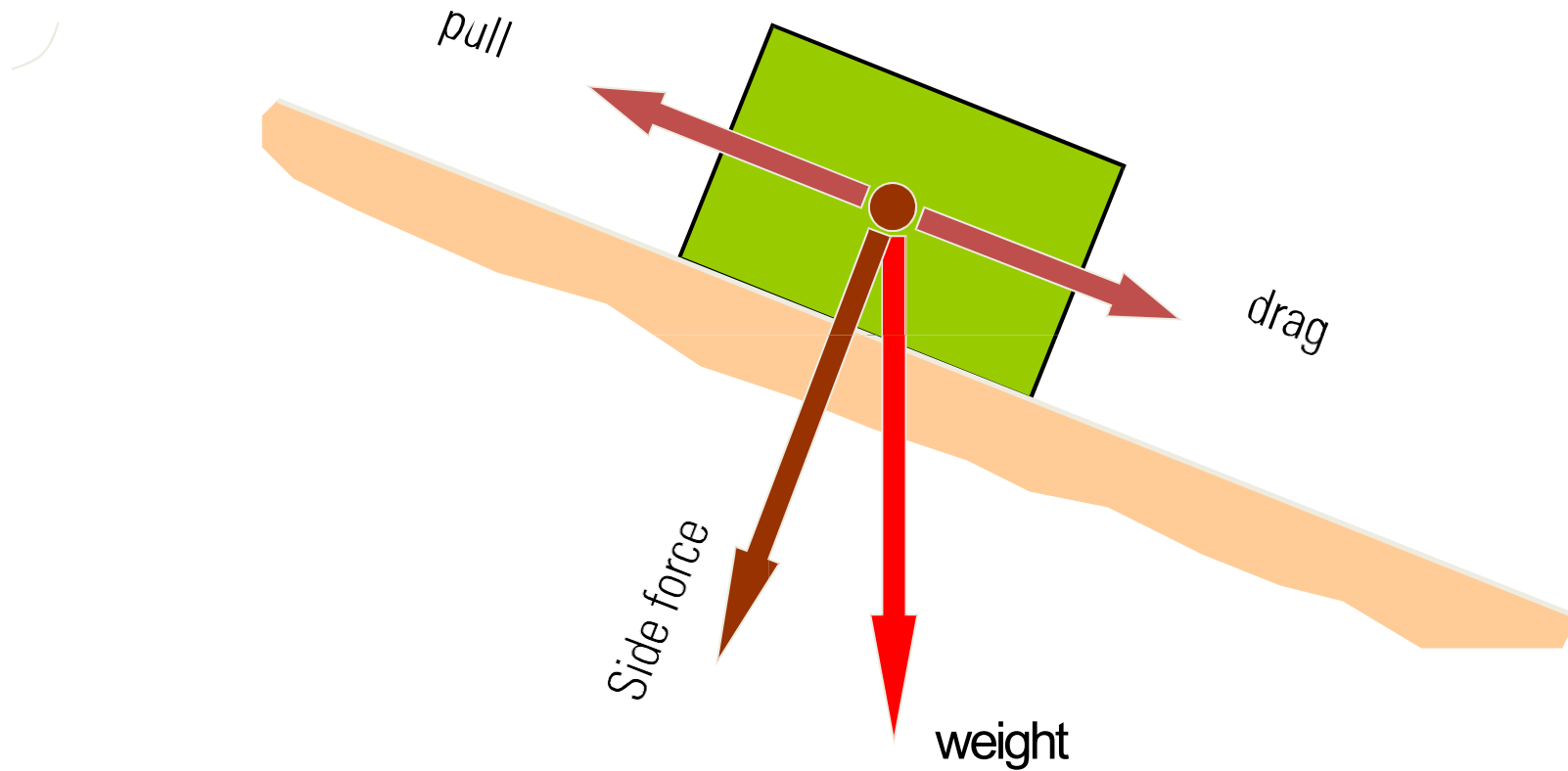
Tension more pronounced with high tension and high dog legs

DLS Magnifies effect of tension

Stiffness negligible effect with dog legs less than 15 deg/100ft

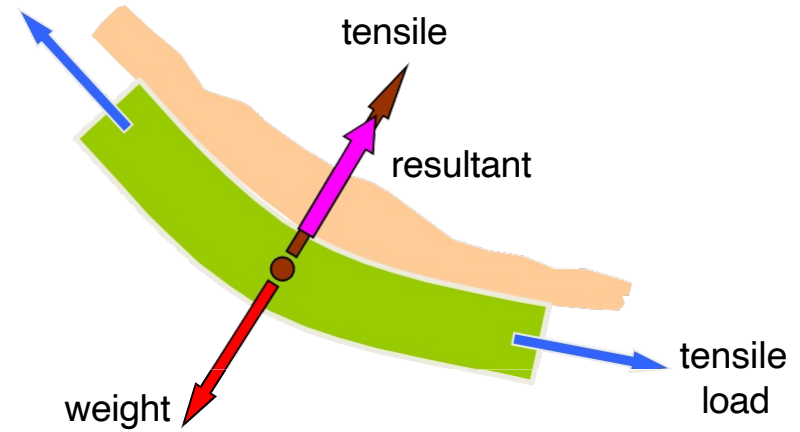
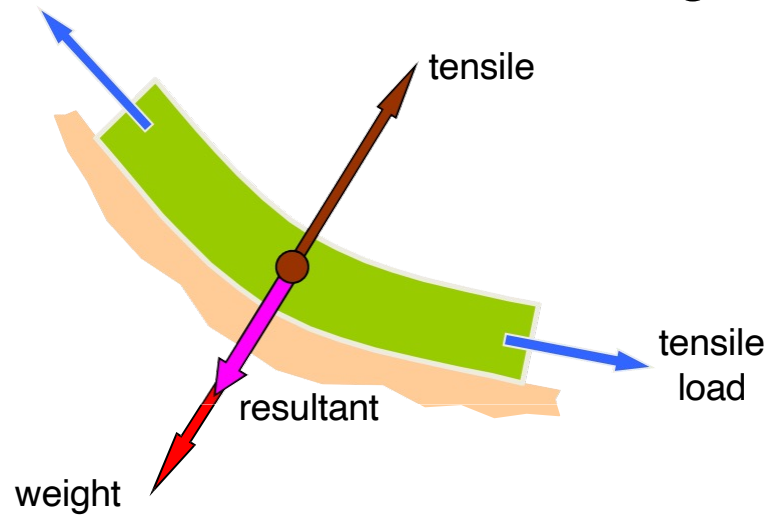
Buckling high compressional loads with neutral point significantly above the bit (near surface)

The Weight Component of Side Force

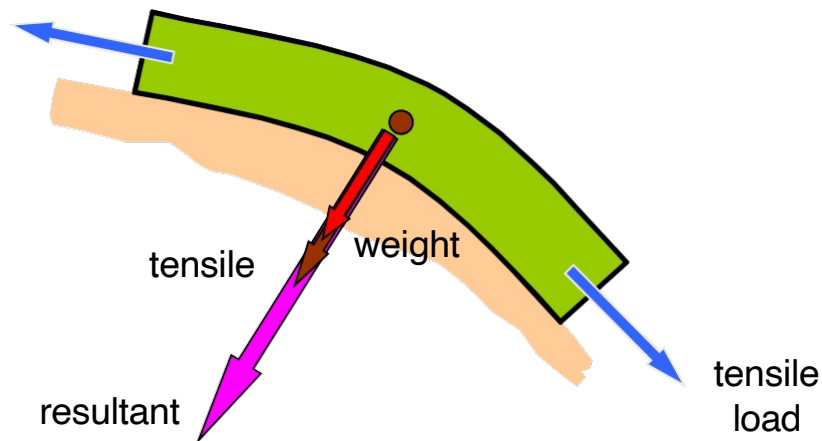


Side Forces – Tension and DLS

Building Section



Dropping Section



Sidewall Forces – Tension and DLS*

Wall force with pipe tension and DLS:

$$SF = \frac{DLS \times \pi \times L \times T}{18 \times 10^3}$$

SF = SideForce (lbf / distance L)

DLS = Dog Leg Severity (Lubinski) (deg/100ft)

L = Length of section (ft)

T = Tension (lbf)

Exercise

Example:

Calculate the wall force across a 30' section of 5°/100' DLS considering a tension of 100,000 lbs below the DL.

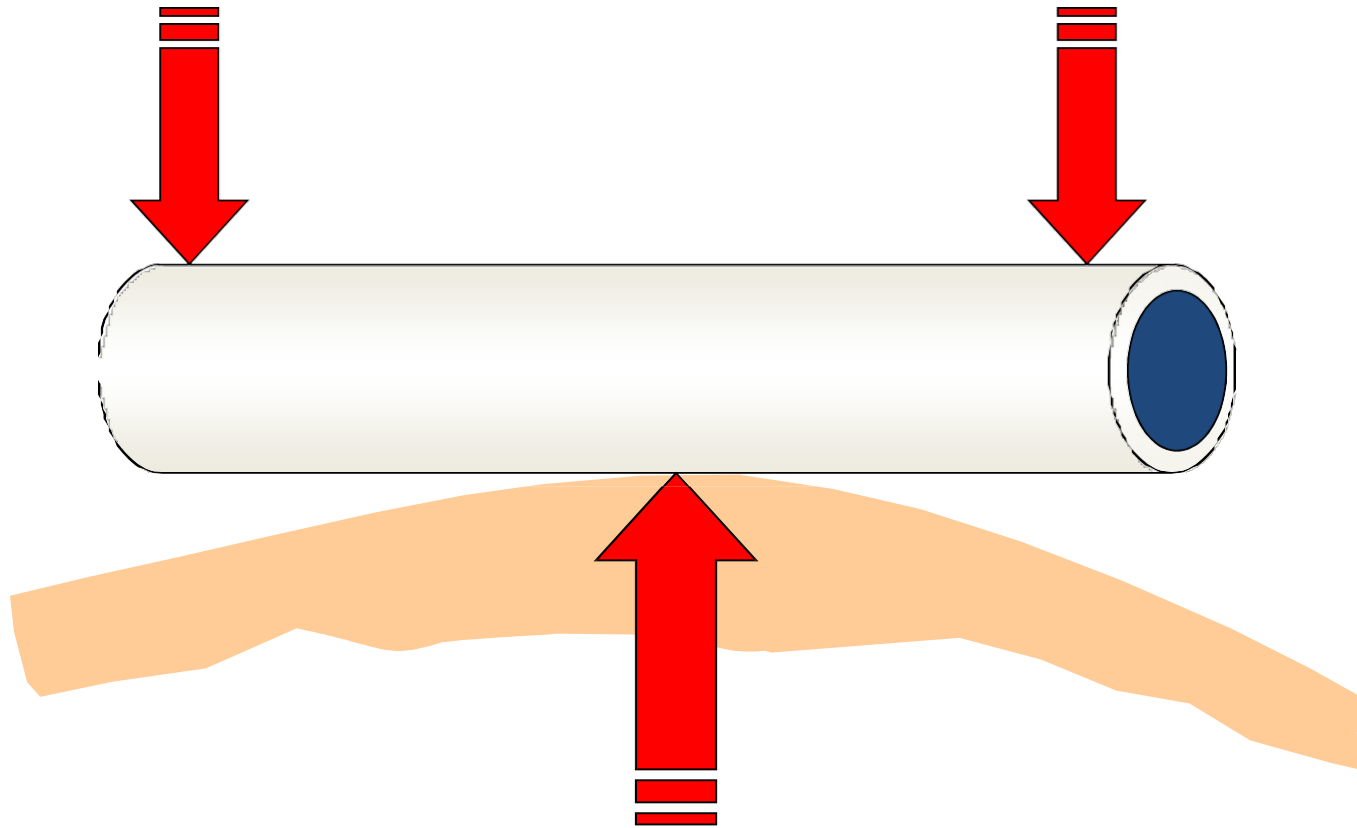
$$SF = \frac{5 \times \pi \times 30 \times 100000}{18 \times 10^3} = 2617.91 \text{ lbf} / 30 \text{ ft}$$

Exercise:

KOP of 1500' and a build up to 30° inclination. Our TD is 10,000'. The drillstring tension at 1500' when we are drilling at TD could be around 180,000 lbs. If the average length of a joint of drillpipe is 31' and if we want to limit our side force to 2,000 lbs per joint of drillpipe what is the maximum DLS can be used?

$$DLS = \frac{18 \times 10^3 \times SF}{\pi \times L \times T} = \frac{18 \times 10^3 \times 2000}{\pi \times 31 \times 180000} = 2.05^\circ / 100 \text{ ft}$$

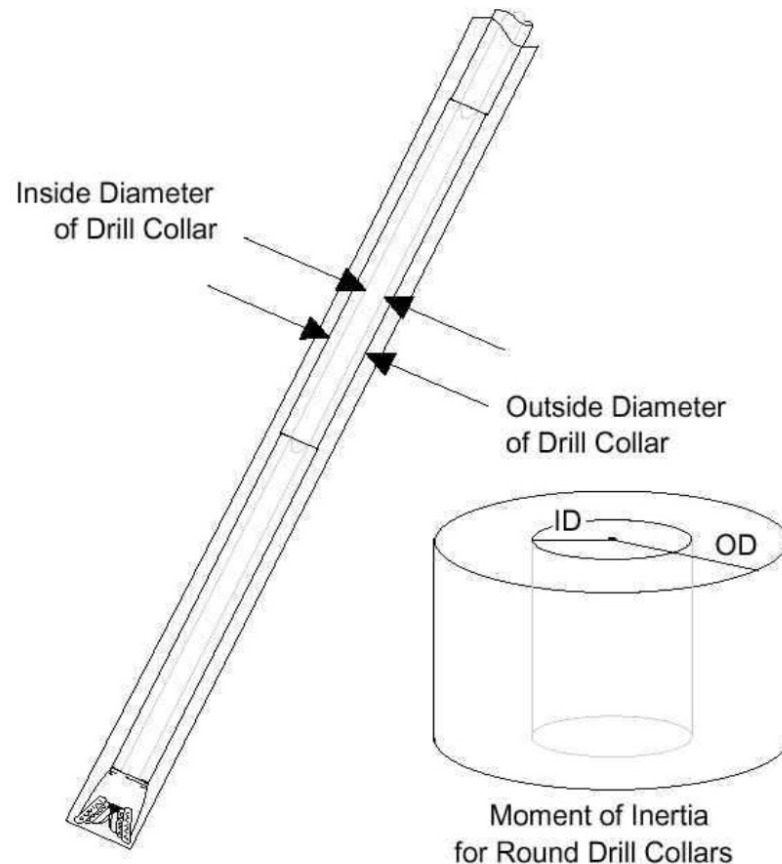
The Stiffness Component of Side Force



When does stiffness start to become a factor?

5" drill pipe	16 deg/100ft
3 1/2" drill pipe	22 deg/100ft

Stiffness – BHA as a Hollow Cylinder



Stiffness Coefficient = $E \times I$

where:

E = Young's Modulus (lb/in²)

I = Moment of Inertia (in⁴)

Moment of Inertia

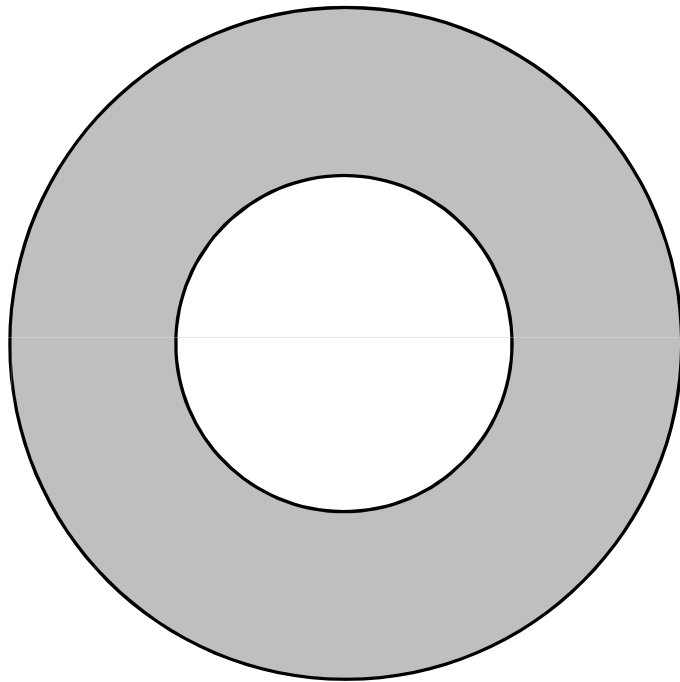
$$I = p (OD^4 - ID^4) \div 64$$

OD = outside diameter

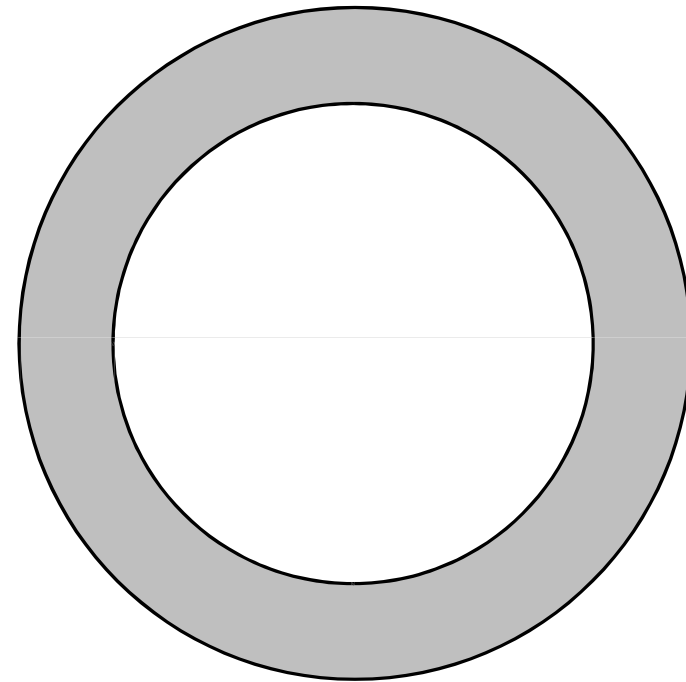
ID = inside diameter

Stiffness – BHA as a Hollow Cylinder

Which one is more stiff?

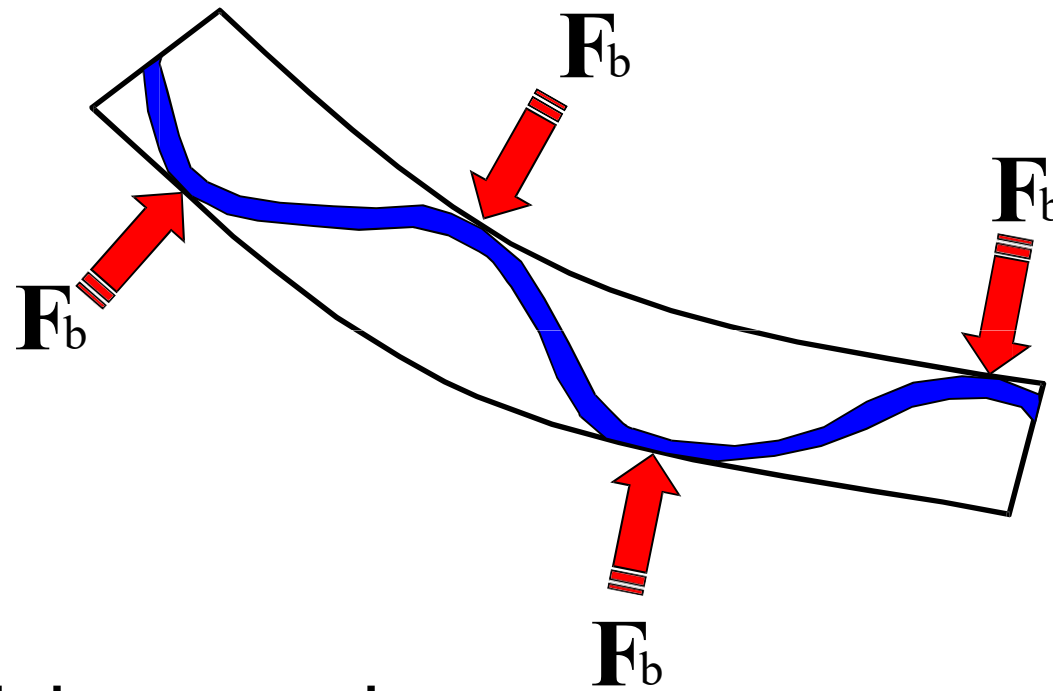


Drill Collar?



Drill
Pipe?
Casing?
Liner?

The Buckling Component of SideForce



String is in compression

Summary

Five Components of Side Force

Weight	always a consideration, use light drill pipe in Horizontal wells
Tension	more pronounced with high tension and high dog legs
DLS	Magnifies effect of tension
Stiffness	negligible effect with dog legs less than 15 deg/100ft
Buckling	high compressional loads with neutral point significantly above the bit (near surface)

Effect of high side forces

1. Increased Friction forces (i.e more torque and drag)
2. Wear => Casing, Drill string components



Recommendation by Aurthur Lubinski: keep sideforces below 2,000lbs/30ft to avoid problems

Torque and Drag Modeling- Outputs

1. Side Forces
2. Axial Loads (Tension, compression, NP location, buckling)
3. Torque
4. Axial Displacement
5. Stresses

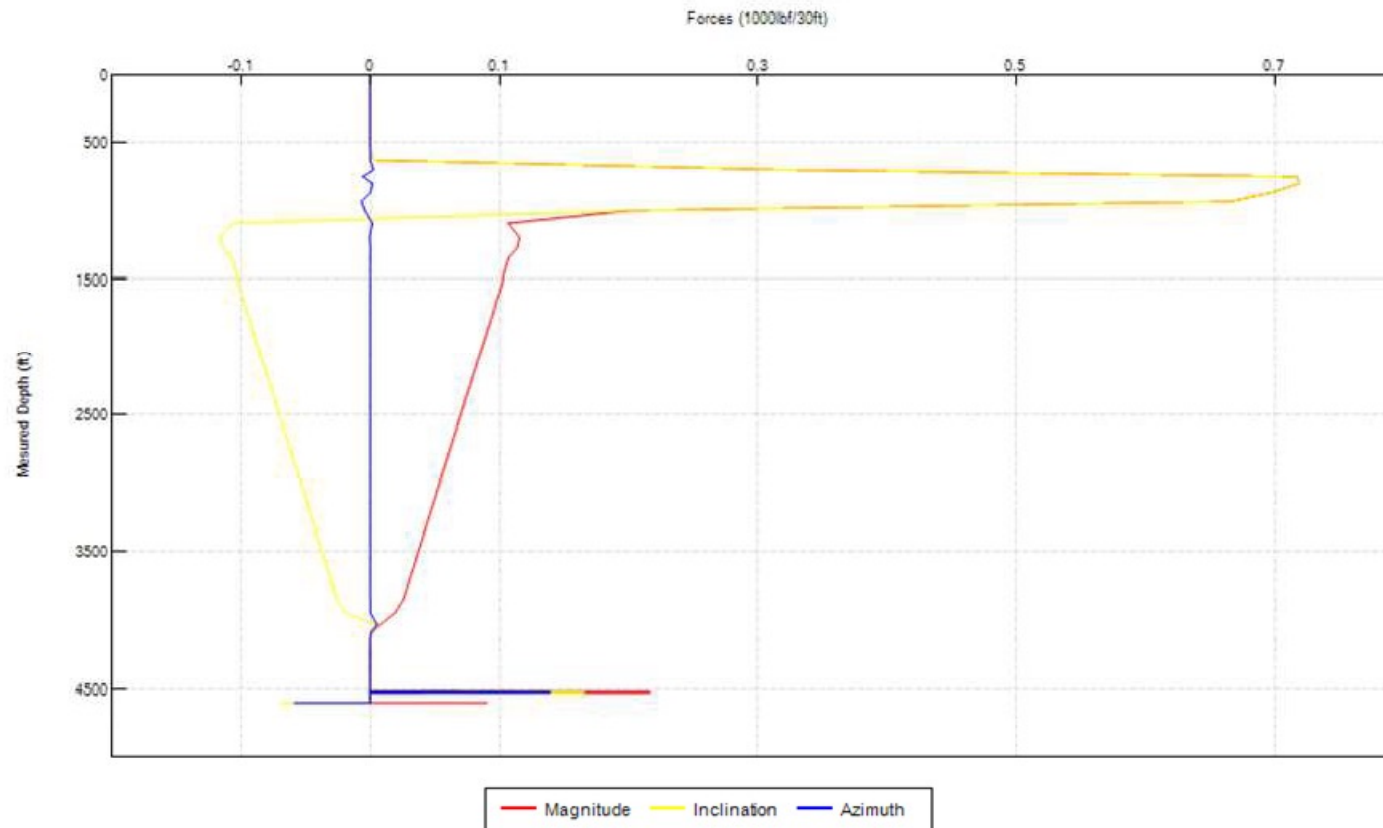
Torque and Drag Modeling- Outputs

Side Forces

Design Objective is to keep less than 2,000 lbs per 30ft

Side Forces

Mudweight = 13 (lbm/gal), DWOB = 30 (1000 lbf), DTOR = 3 (1000 ft.lbf)
BlockWeight = 90 (1000 lbf), Bit Depth = 4603 (ft)
Well: Sinan Well 09, Borehole: Sinan Well 09
Client: Alistair Oil
Scenario : Drilling Scenario, Date: Mar 02, 2010



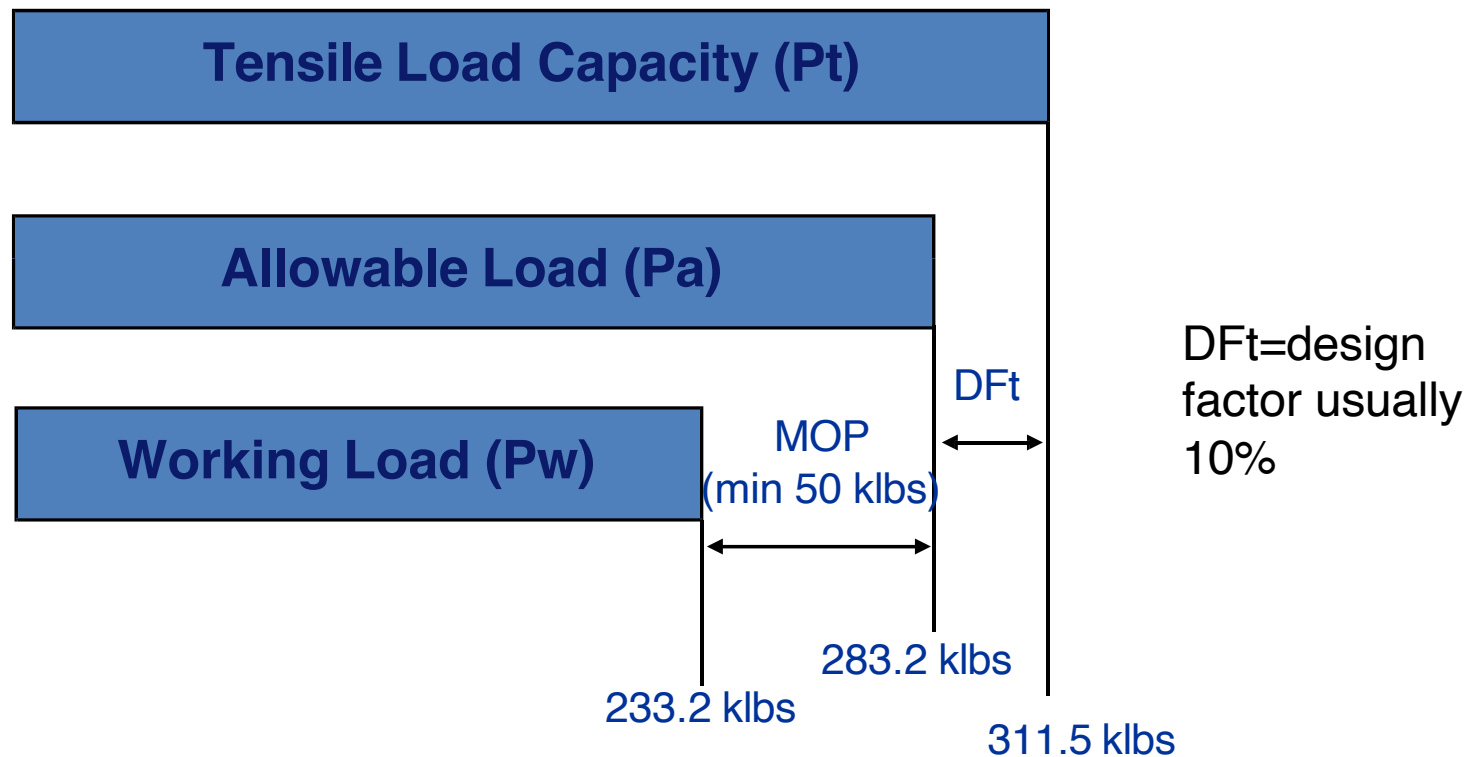
Torque and Drag Modeling- Outputs

Axial Loads

Design Objective is 1. to provide adequate margin of overpull without exceeding 90% tensile capacity and
2. to avoid buckling when in compression

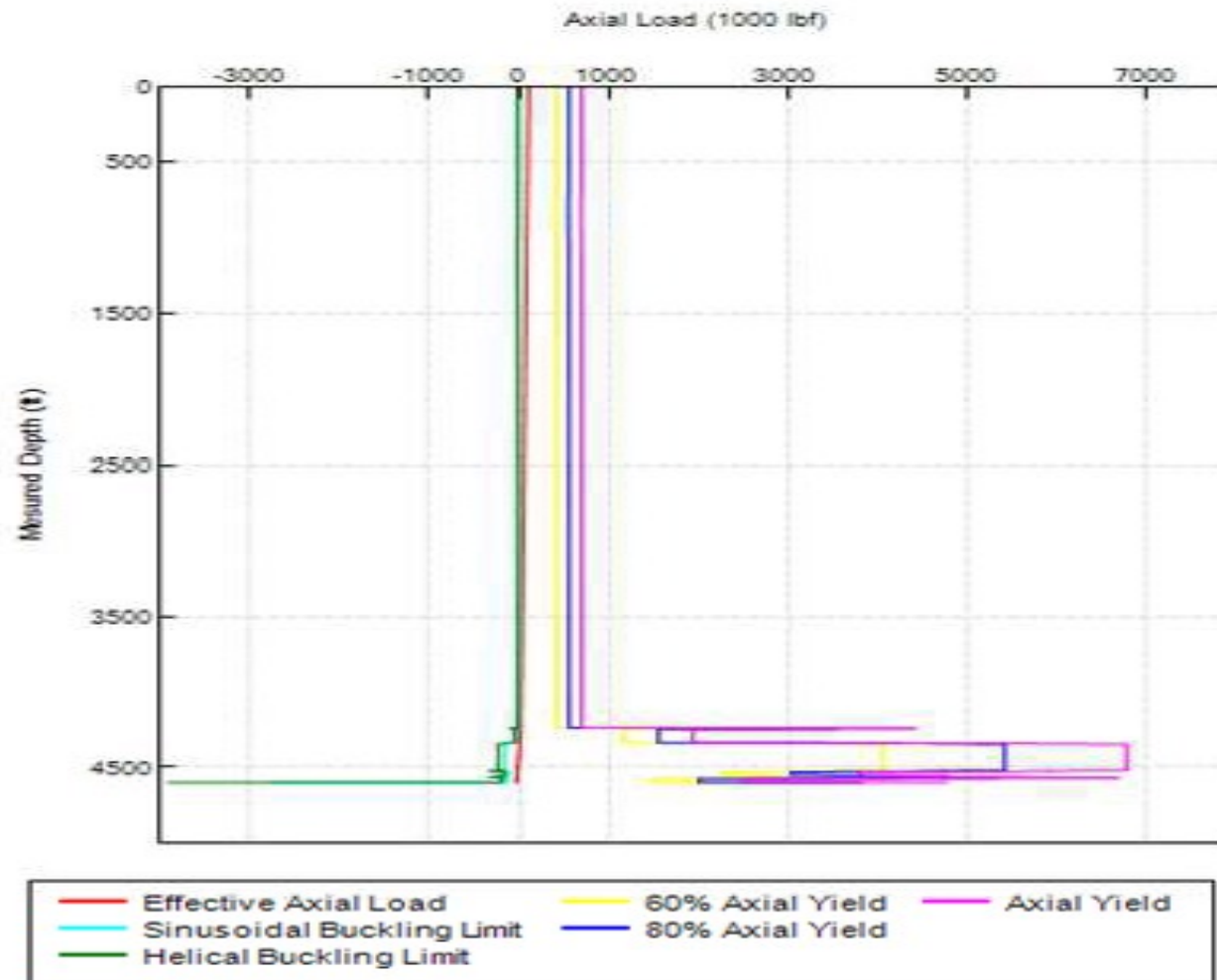
Axial loads -Tensile Capacity

5" 19.5 lb/ft Grade E, premium Class DP



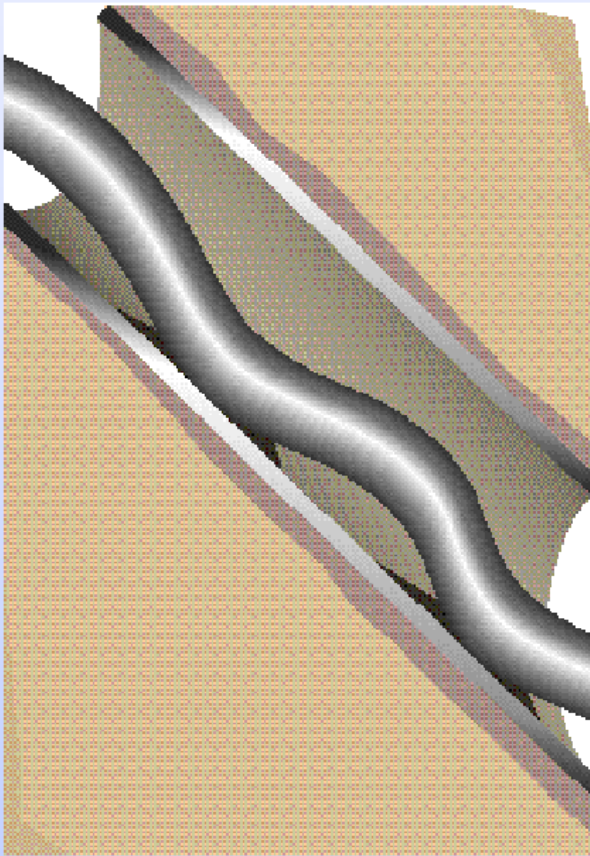
Axial Load

Mudweight = 13 (lbm/gal), DWOB = 30 (1000 lbf), DTOR = 3 (1000 ft.lbf)
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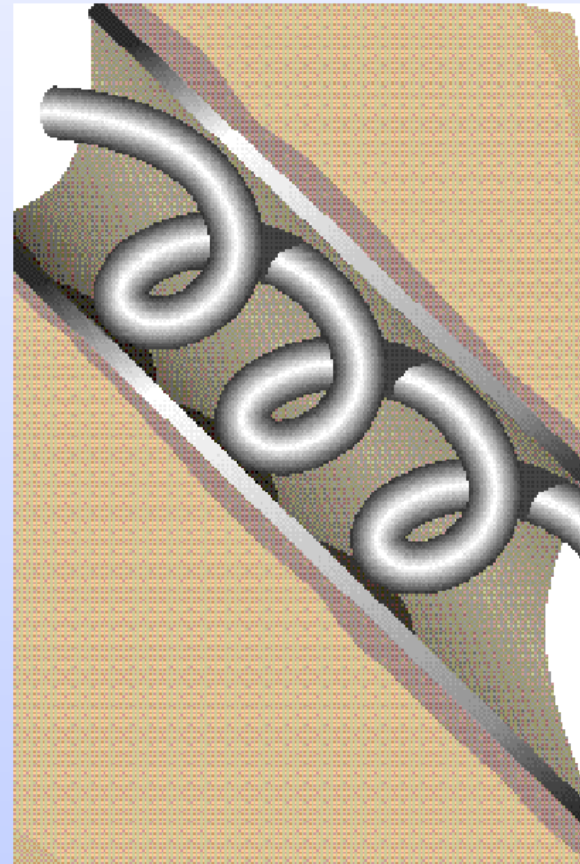
Sinusoidal & Helical Buckling

SINUSOIDAL BUCKLING



WOB Critical

HELICAL BUCKLING



WOB Helical

WOB →



Dawson-Paslay Buckling Criteria

$$F_{CR} = 2 \times \sqrt{\frac{E \times I \times K_B \times W \times \sin \theta}{r}}$$

F_{CR} = Critical sinusoidal buckling load (lbs)

θ = Inclination of the hole at the point of interest (deg)

E = Young's Modulus

K_B = Buoyancy factor (unitless)

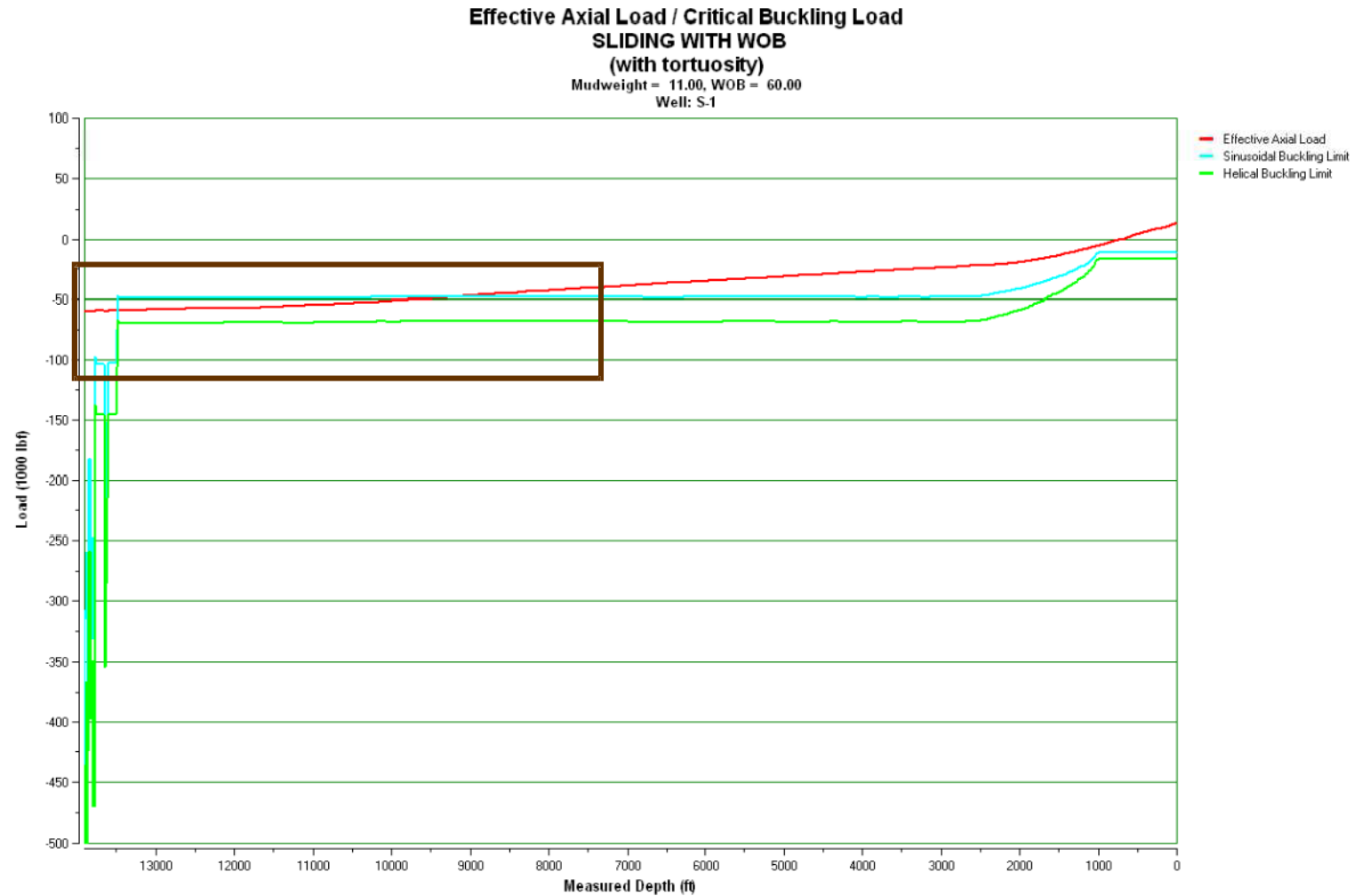
I = Moment of inertia (inch⁴)

W = Unit weight in air (lbs/in)

r = Radial clearance between pipe tool joint and hole (in)

Thus buckling more likely at lower Inclination

Buckling - Worst Case Scenario???



Torque and Drag Modeling- Outputs

Torque

Design Objective is to keep drill stem torque less than 80% of torsional yield and less than 80% of make up torque (MUT)

Stress Yet?

Stress-Strain relationship

Hooke's Law states that;

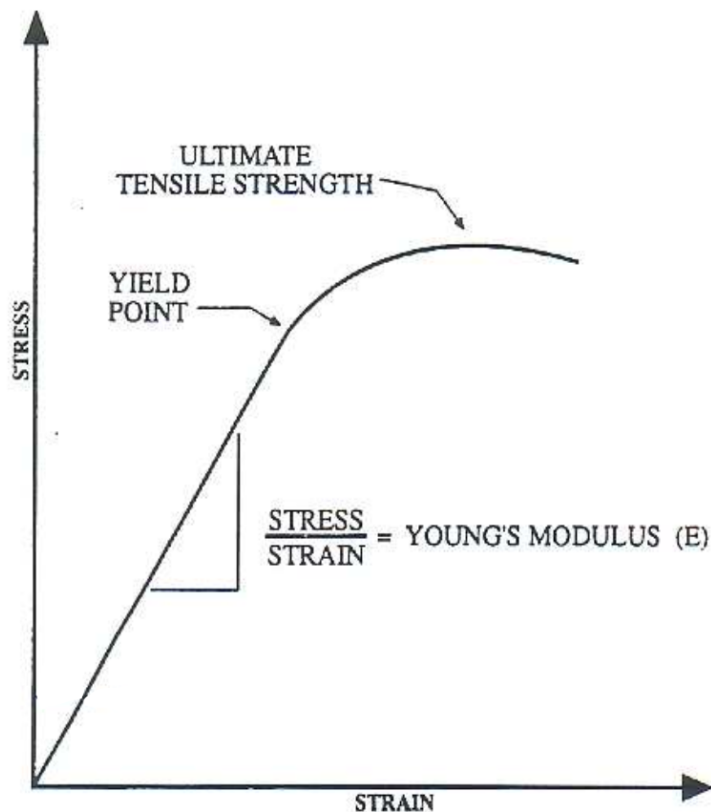
“Within the Elastic Limit, Stress is proportional to Strain”.

If $\text{Stress} \propto \text{Strain}$, then $\text{Stress} \div \text{Strain}$ must be a constant.

This constant is called Young's Modulus of Elasticity. The Greek symbol E (Epsilon) is used to denote Young's Modulus.

E for Steel = 30,000,000 psi
(30×10^6 psi)

E for Aluminum = 10,500,000 psi
(10.5×10^6 psi)



Yield Strength / Tensile Strength

- Yield Strength:

The level at which the material changes from predominately elastic to predominately plastic strain behavior. Unit for this measure is PSI.

- Tensile Strength: is the highest stress level a material achieves before it breaks.

Drillpipe Strength

Drillpipe grade determines minimum yield strength

- E 75,000 psi
- G 105,000 psi
- S 135,000 psi
- Z 140,000 psi
- V 150,000 psi

Connection strengths are not related to the pipe body grade Drillpipe body is most likely to be the tensile failure limit Drillpipe tool joint is most likely to be the torsional failure limit.

Premium connections improve torsional limits and the drillpipe body becomes the torsional limit

Keep drilling/reaming torque < 80% of MUT

NOTE:

All API tool joints have a minimum yield stress of 120,000 psi regardless of the grade of the drill pipe they are used on (E, X, G, S).

Classes of Drillpipe

American Petroleum Institute Standards

- RP 7G with drillpipe and collars and RP5 for casing/tubing
- Most specifications sheets will state premium
- Most marketing sheets specify “new” values

Drill Pipe Classification

- Size: 2 3/8" to 6 5/8" – refers to OD of pipe body
- Length: Range 1: 18-22 ft, Range 2: 27-30ft, Range 3: 38-45 ft
- Grade: E - 75, X - 95, G - 105, S - 135
the numbers denote 1000's of psi minimum yield stress
each grade has a min, max and average yield stress
- Weight: Called nominal weight
- Class: API classification for used pipe

For example a drill pipe could be: 5", Range 2, G-105, 19.5ppf, New

CLASS

Drill Pipe

New Pipe
Premium

Original Dimensions.

Diameter reduction not higher than 3% of OD
Wall thickness not less than 80%

Class 2

Diameter reduction not higher than 4% of OD
Wall thickness not less than 70%

Class 3

Other imperfection that exceeds Class 2

Color codes of Drill Pipe

- 2 white bands
- 1 white band
- 1 yellow band
- 1 Orange band
- 1 Red band on tube
- 1 Red band on tool Joint

Tool joint premium

New Pipe

Tool Joint Class 2

Class 3

Junk

Junk and can be
repaired



Stress

A point within a body under loading can be subjected to FOUR possible types of stresses:

- NORMAL STRESS,
- BENDING STRESS,
- SHEAR STRESS,
- TORSIONAL STRESS

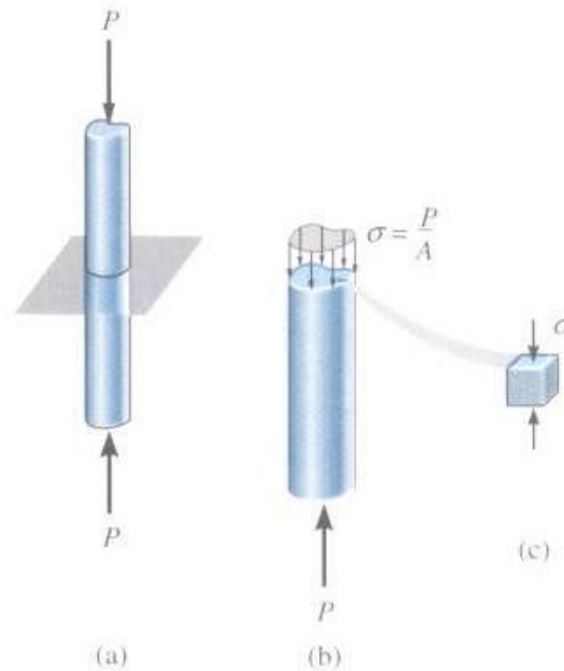
The magnitude of these stresses is dependent on the loading conditions of the body of interest.

Normal Stress

Normal Stress is the intensity of the net forces acting **normal (perpendicular)** to an infinitely small area A within an object per unit area.

If the normal stress acting on A pulls on it, then it is referred to as **tensile stress**,
If it pushes on the area, it is called **compressive stress**.

$$\sigma = \frac{P}{A}$$



Bending Stress

Bending Stress

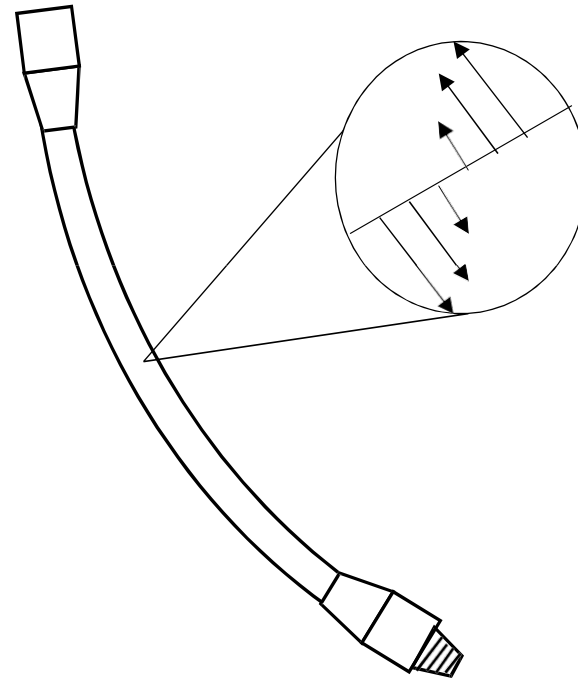
$$\sigma_b = \frac{E \times D}{2R}$$

(*)

E = Young's Modulus (psi)

D = Diameter of the Tubular (inches)

R = Radius of Curvature (inches)



(*)

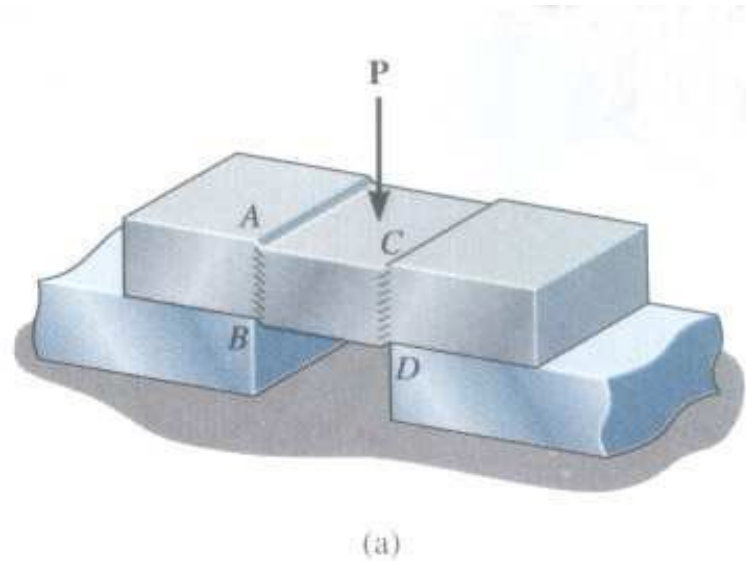
SPE 37353

Drill-Pipe Bending and Fatigue in Rotary Drilling of Horizontal Wells - Jiang Wu

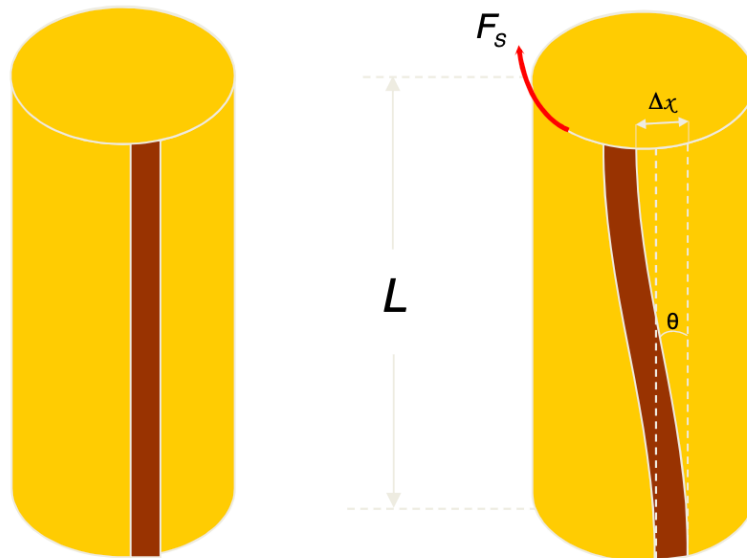
Shear Stress

Shear Stress is the intensity of force per unit area, acting tangent to A.

If the supports are considered rigid, and P is large enough, the material of the bar will deform and fail along the planes AB and CD.



Torsional Stress



$$\frac{\text{Shear Stress}}{\text{Shear Strain}} = \frac{F_s / A}{\theta} = G \text{ (Shear Modulus)}$$

$$\sigma_{\tau} = \frac{\pi \times G \times d \times N}{12 \times L}$$

or

$$\sigma_{\tau} = 6 \times d \times \frac{Q}{J} \quad \text{Where } Q = \frac{\pi \times G \times J \times N}{72 \times L}$$

$G \rightarrow$ Shear Modulus of steel, 12×10^6 psi

$N \rightarrow$ Number of turns in the drill pipe string, rev

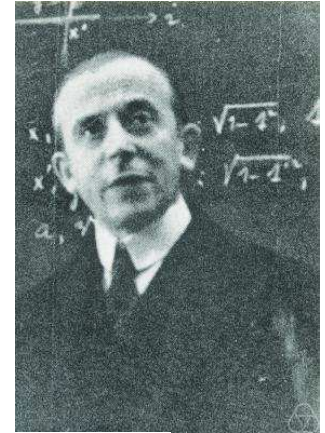
$Q \rightarrow$ Torque applied to the DP, ft.lb

$L \rightarrow$ Length of Drillpipe string, ft

$d \rightarrow$ Internal diameter of the pipe, inches

$J \rightarrow$ Polar moment of inertia, $\frac{\pi}{32} \times (D^4 - d^4)$; inch⁴

Richard Von Mises



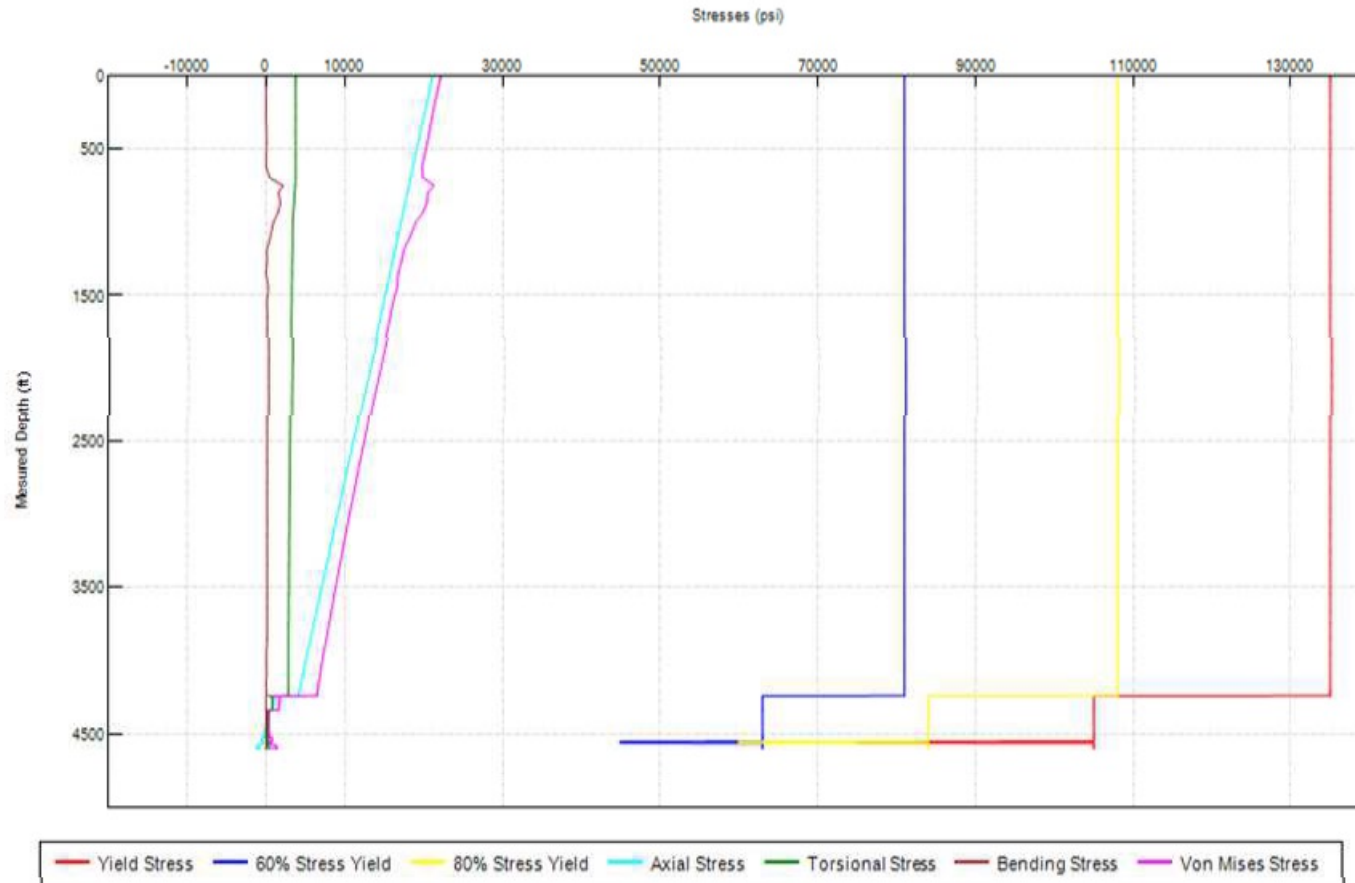
$$\text{Von Mises Stress} = \sqrt{\left(\left(\sigma_{axial}\right) + \left(\sigma_{bending}\right)\right)^2 + 3\sigma_{torsional}^2}$$

- Axial, Bending and Torsional Stresses combined
- Total Stress of the drillstring component [psi]

Design Objective is to keep VM stresses less than 80% of minimum Yield Strength to avoid failure and less than 60% to avoid fatigue

Stresses

Mudweight = 13 (lbm/gal), DWOB = 30 (1000 lbf), DTOR = 3 (1000 ft.lbf)
BlockWeight = 90 (1000 lbf), Bit Depth = 4603 (ft)
Well: Sinan Well09, Borehole: Sinan Well09
Client: Alistair Oil
Scenario : Drilling Scenario, Date: Mar 02, 2010



Torque & Drag

Definition & Monitoring

Torque and Drag - Definition

Torque Losses

Are defined as the difference between the torque applied at the rig floor and the torque generated at the bit. Also referred to as rotating friction.

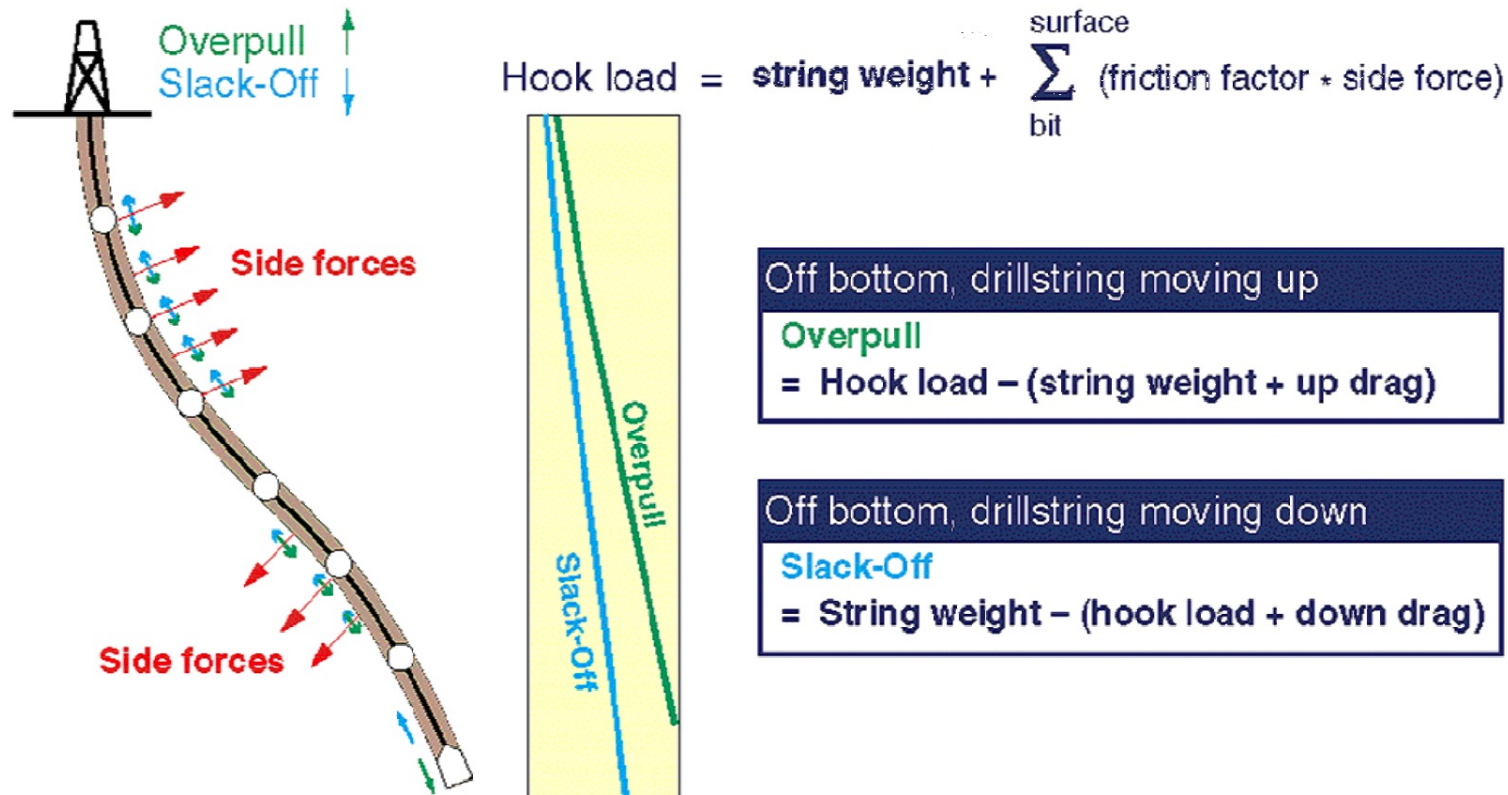
Drag losses

It is the difference between the static weight of the drillstring and the weight under movement. Also referred to as sliding friction.

drag = sideforce x friction factor

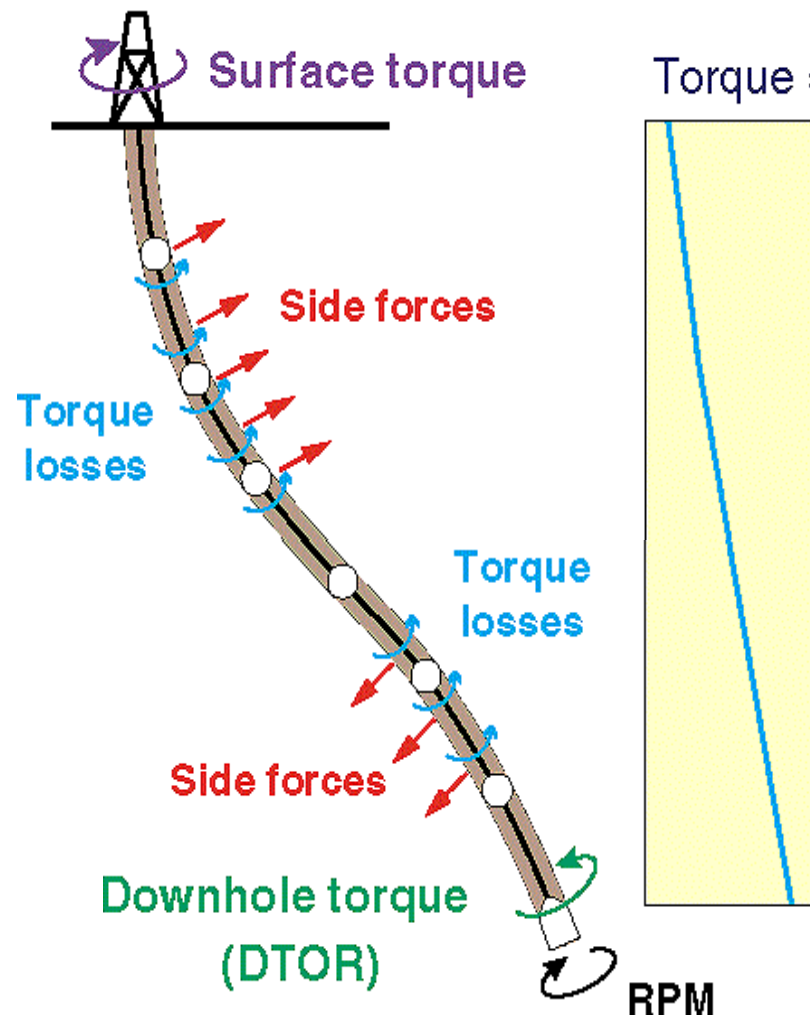
torque = sideforce x friction factor x radius

Overpull / Slack-Off



Overpull: The excess hookload above the expected value (up weight) when picking the string up
Slack Off: The excess reduction in hookload below the expected value (down weight) when running the string in hole.

Torque



$$\text{Torque} = \sum_{\text{bit}}^{\text{surface}} (\text{friction factor} * \text{side force} * \text{element radius})$$

Off bottom, rotating

Excess Torque

$$= \text{Surface torque} - \text{Expected Torque}$$

On bottom

Excess Torque

$$= \text{Surface torque} - (\text{DTOR} + \text{Expected Torque})$$

Torque and Drag Monitoring Why

- Track hole condition and deterioration
- Determine hole cleaning efficiency
- Evaluate cuttings bed formation
- Determine limitation of equipment and maximum achievable depths
- Determine mud lubricity effects
- Determine effects of mud weight and mud property changes
- Build a friction factor database
- Understand problems encountered when running casing/liners
- Optimize string configurations and BHA and need for torque reducers

Parameters to monitor

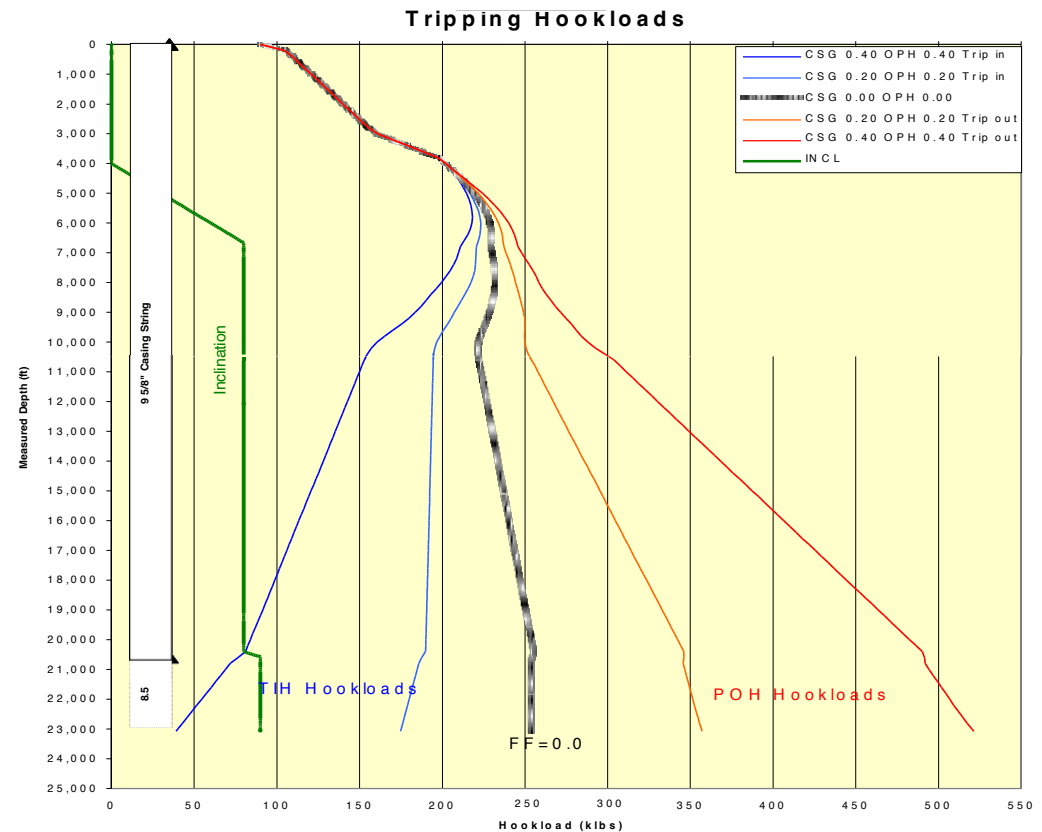
A total of 4 measurements required to monitor T&D

Hookloads

- Picking Up
 - at least 5-6 meters with a constant speed
- Slacking Off
 - at least 5-6 meters movement with a constant speed
- Rotating off bottom
 - at least 1-2 meters off bottom

Torque

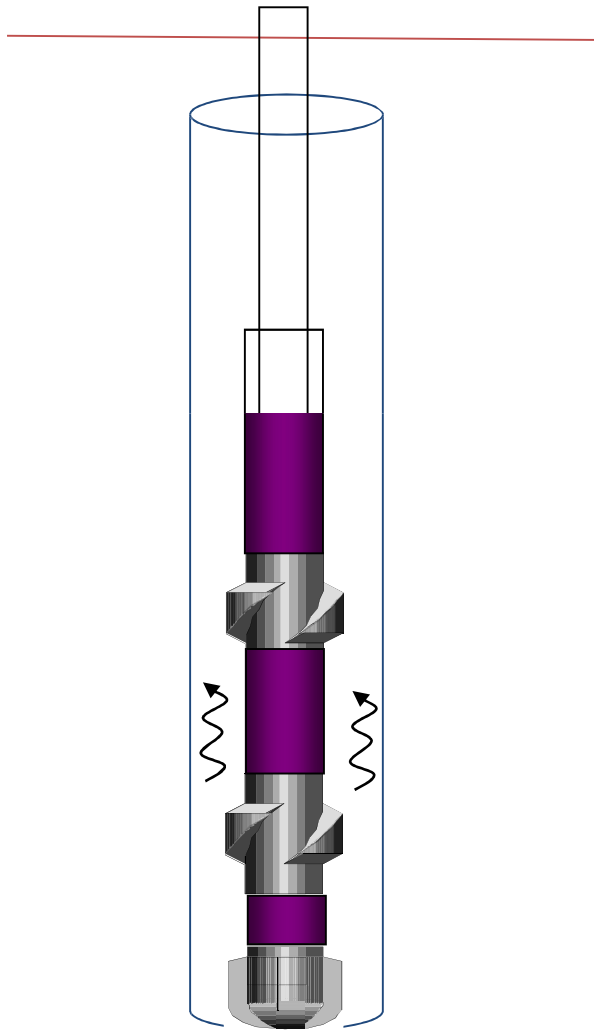
- Off bottom torque @ rotary speed



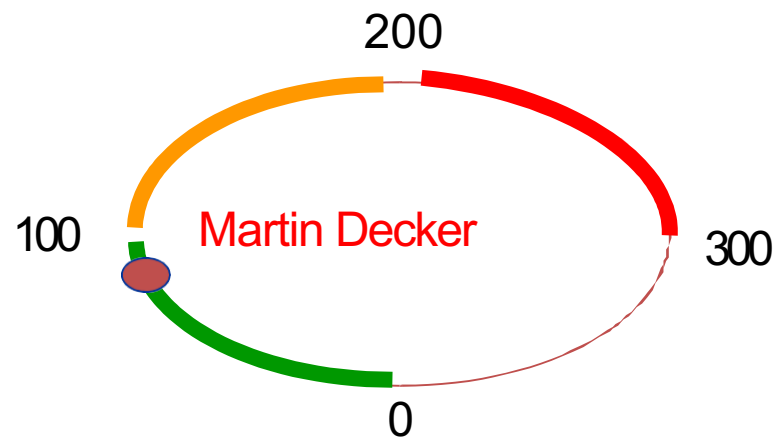
Torque and Drag Monitoring When

- **At every connection**
- **While tripping in/out**
- Prior to drilling out/going back into open hole
- After major inclination and azimuth changes
- Before, during and after wiper trips
- Before and after circulating bottoms up and pumping sweeps
- After a mud type change and major mud properties change
- Before and after additions of torque reducers
- At TD before and after hole has been cleaned
- In case of running casing, monitor drag values every 3-5 joints

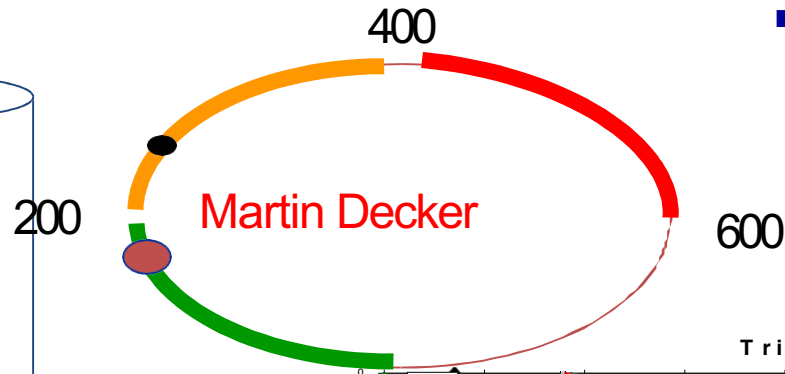
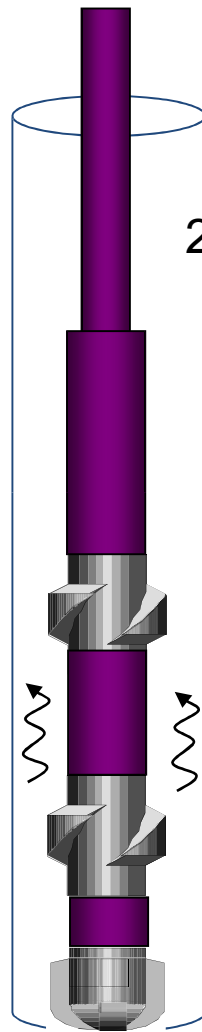
Torque and Drag Monitoring



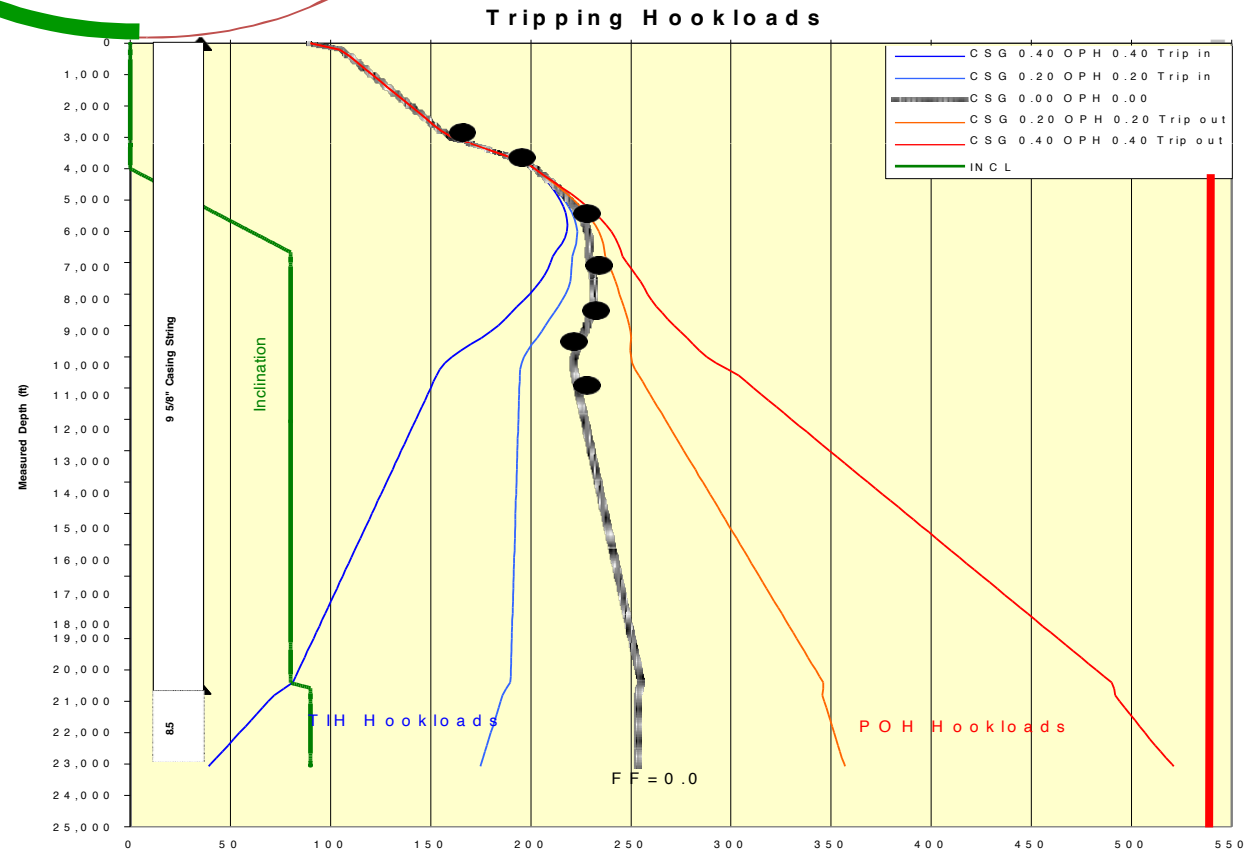
- After drilling down each connection, reciprocate the stand with good circulation and rotation to ensure good hole cleaning and any cuttings are clear of the BHA and to determine if the hole is “free” (situation may be different for different rigs/company procedures, so at each connection, pump/ream the last stand as necessary and as per instructions, for each hole size, angle, formation type, etc).



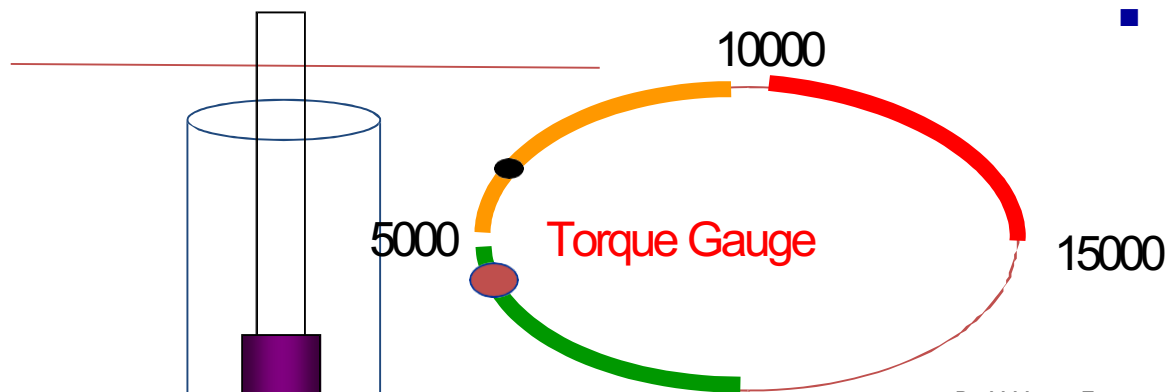
Torque and Drag Monitoring



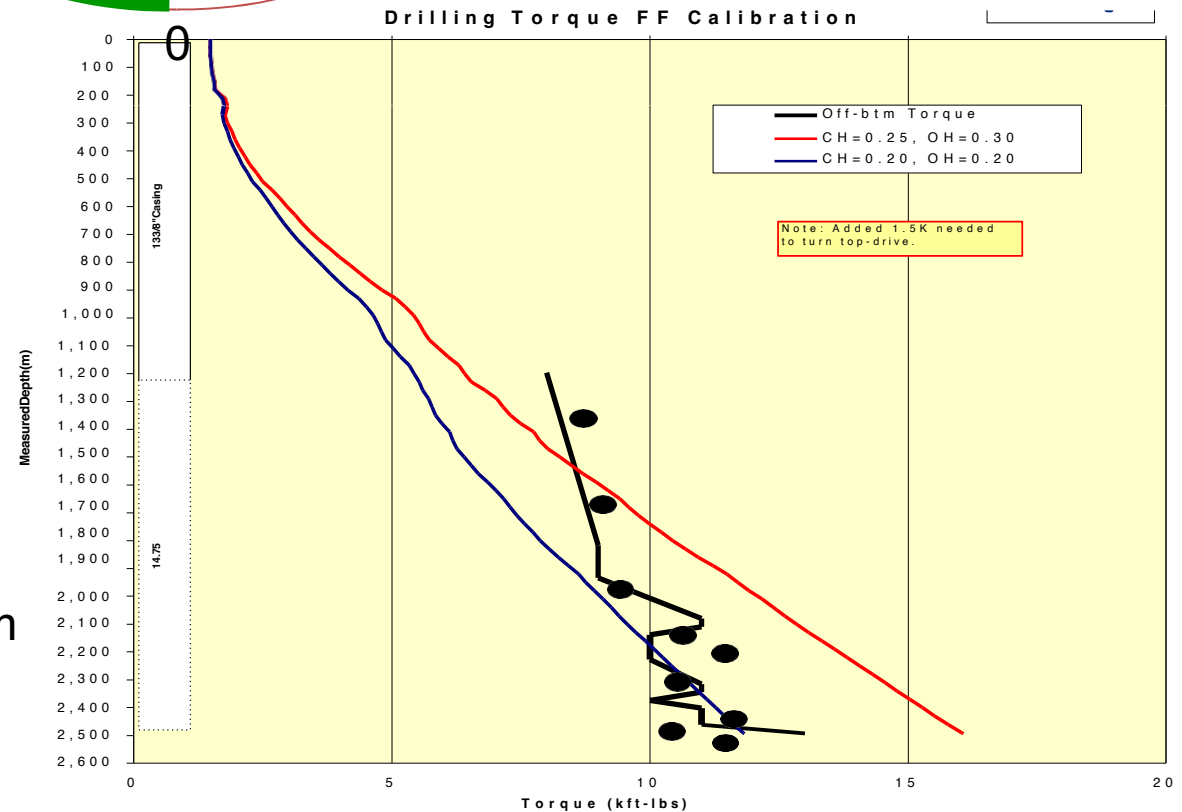
- A few meters off bottom, obtain rotating string weight and torque at drilling RPM and flow rate. If the T&D modeling is done correctly, this weight should be on top of the FF=0 line



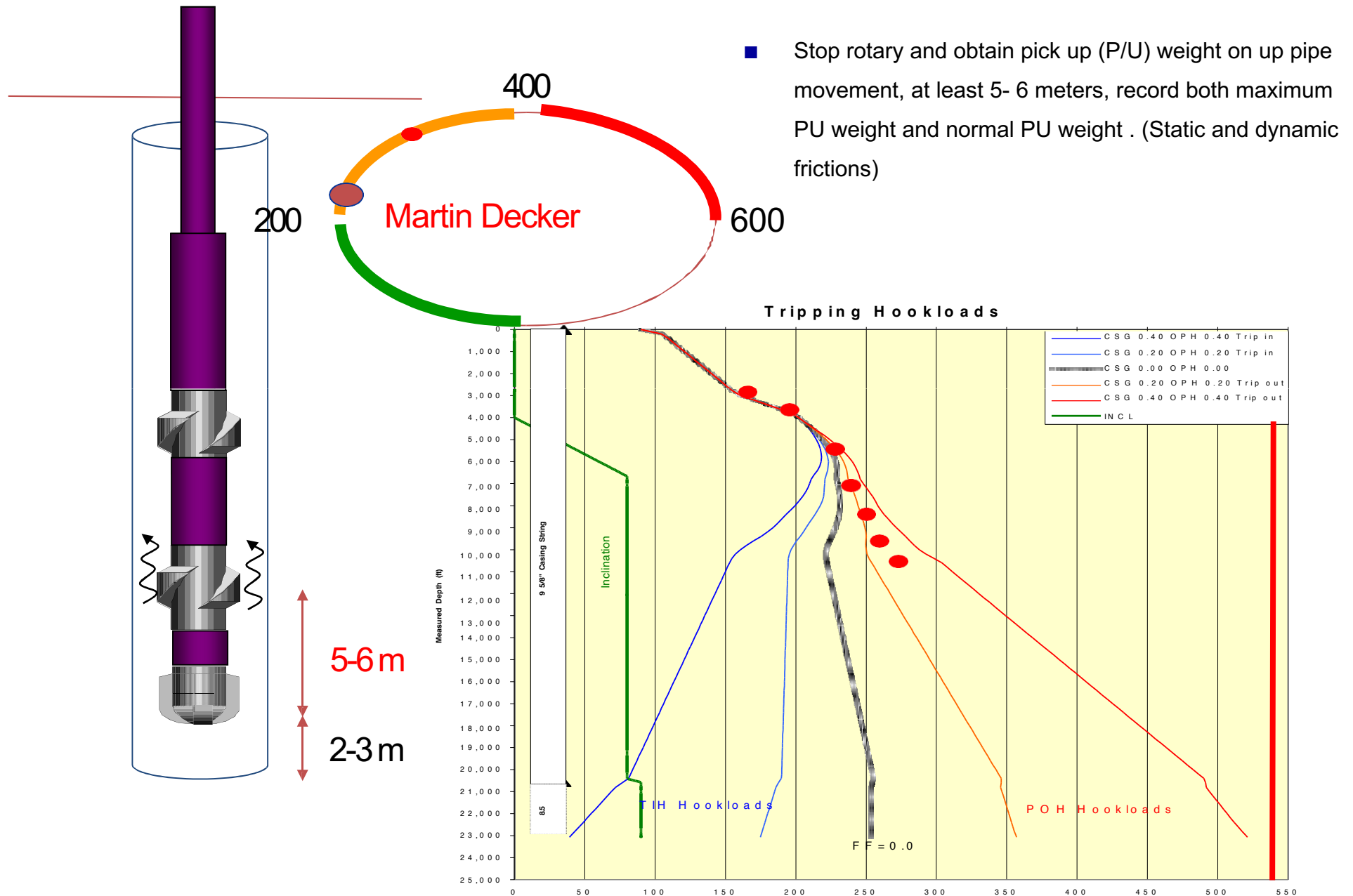
Torque and Drag Monitoring



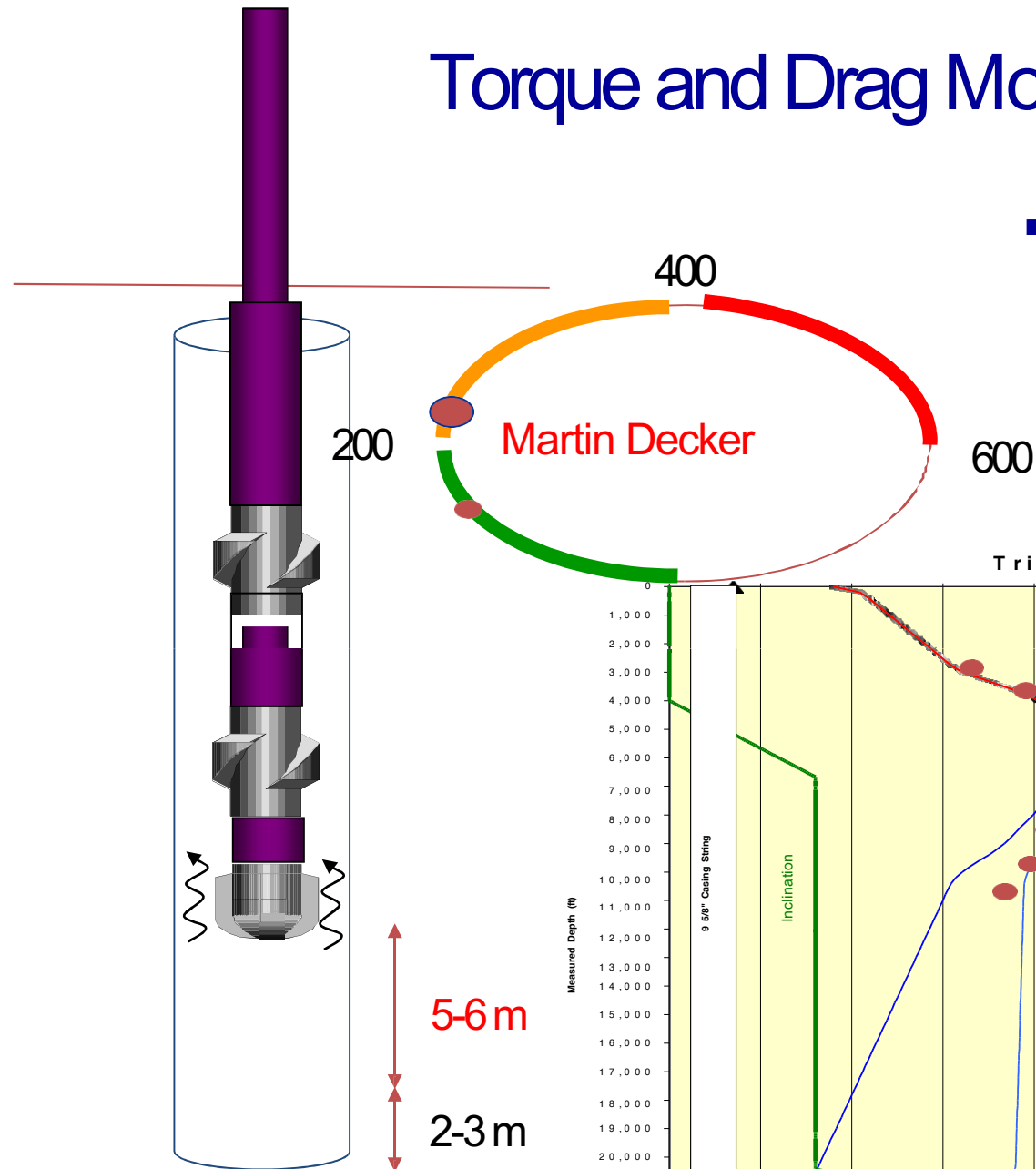
- A few meters off bottom, obtain rotating string weight and torque at drilling RPM and flow rate. If the T&D modeling is done correctly, this weight should be on top of the FF=0 line



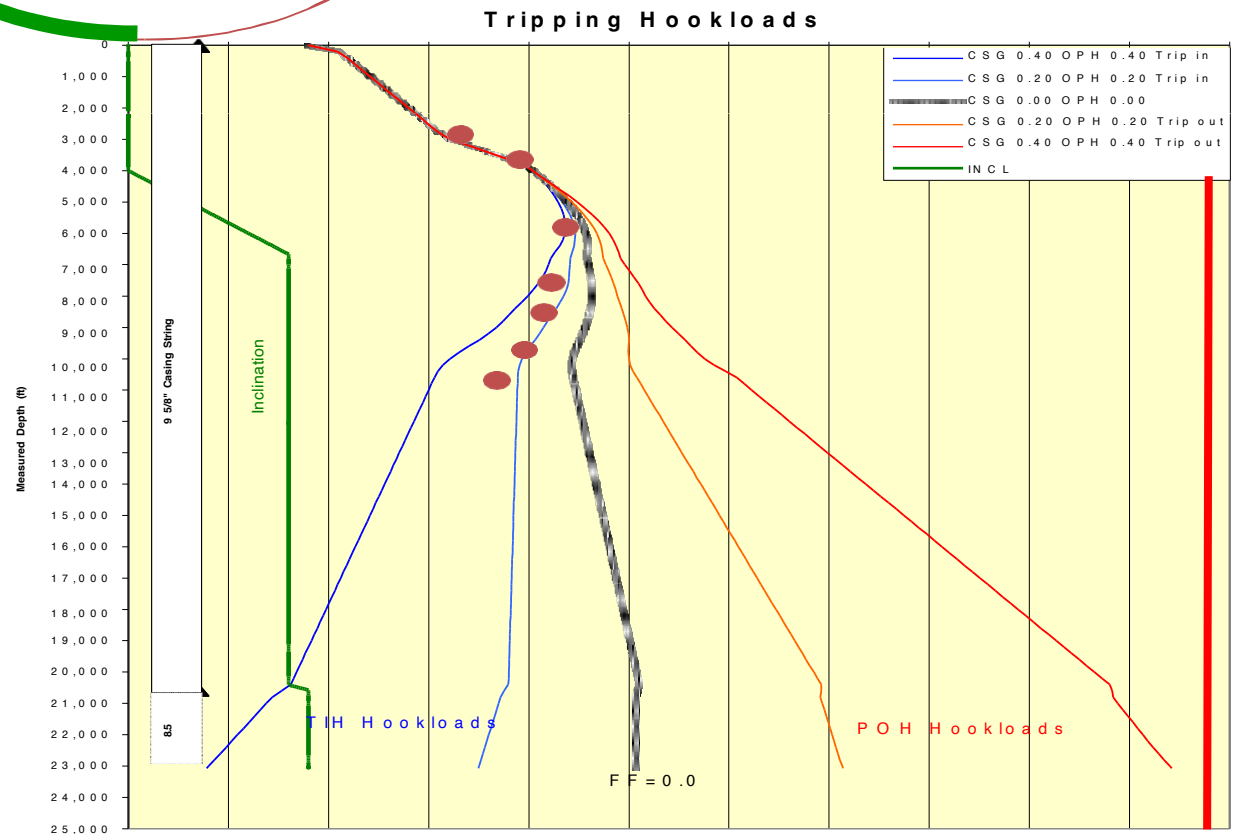
Torque and Drag Monitoring



Torque and Drag Monitoring



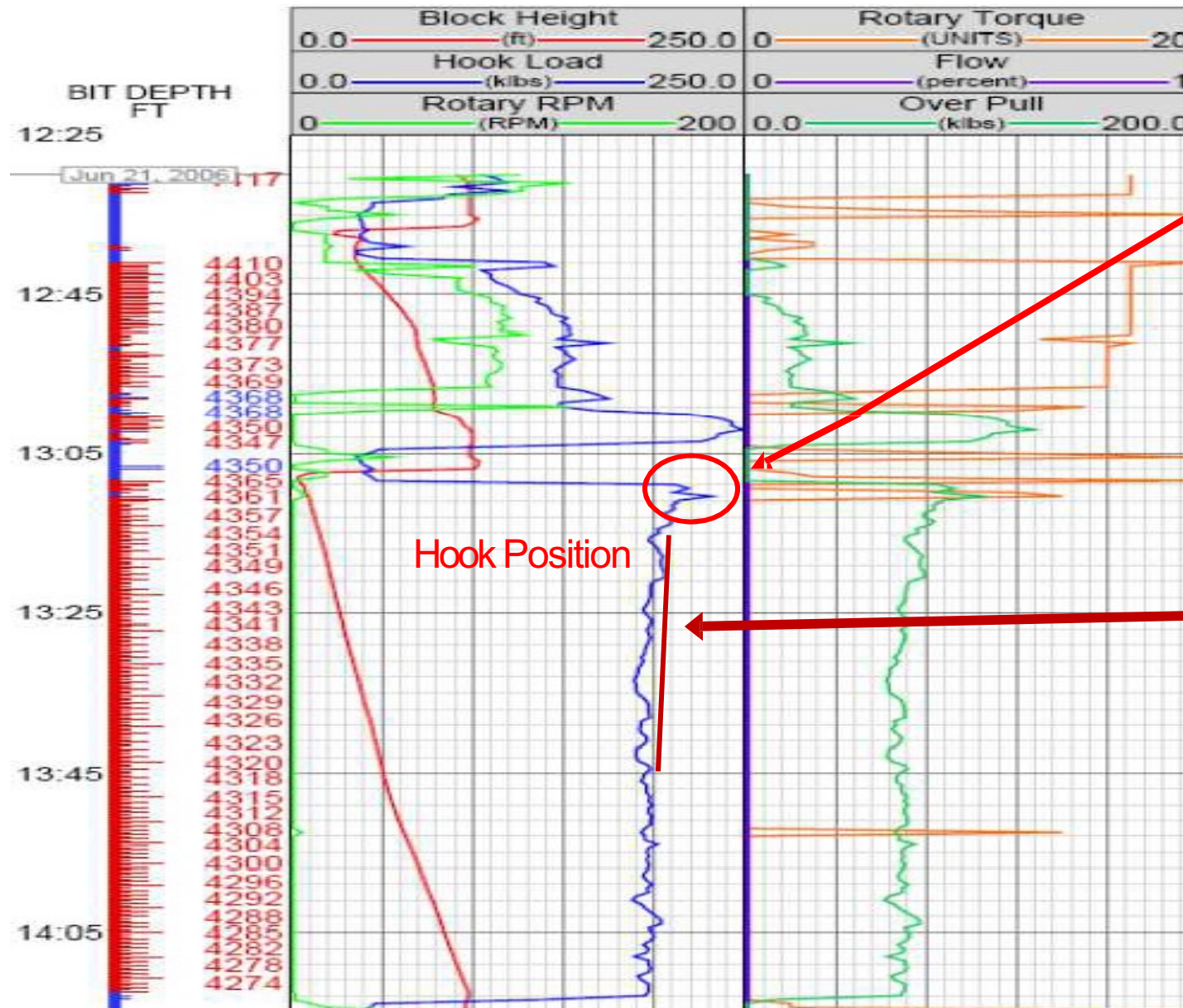
- Obtain the slack off (S/O) weight on the down movement of the pipe while returning the pipe 5-6 meters to bottom. Record both minimum slack off and normal slack off weights.



Torque and Drag Monitoring **How**

- Moving the drill string at the same speed
- Take the least affected, steady weight indicator reading
- Turn pumps off and take P/U and S/O weights and repeat previous steps above, before the connection
- Take the circulating readings at the same flow rate (for each hole section) to avoid the potential influence/interference of hydraulic lift.
- While tripping out, just obtain the pick-up weights. Obtain the slack-off weights while running in.
- Pumps on readings can be used to estimate maximum depth achievable while drilling
- For running casing/liner, get the S/O weights while running.

Typical Hookload Behavior (POOH)



Picking up off the slips, maximum hookload (this represents the static friction factor). This will help us monitor if we are getting closer to rig limits

Steady hookload while moving the drill string up (This represents the dynamic friction factor).

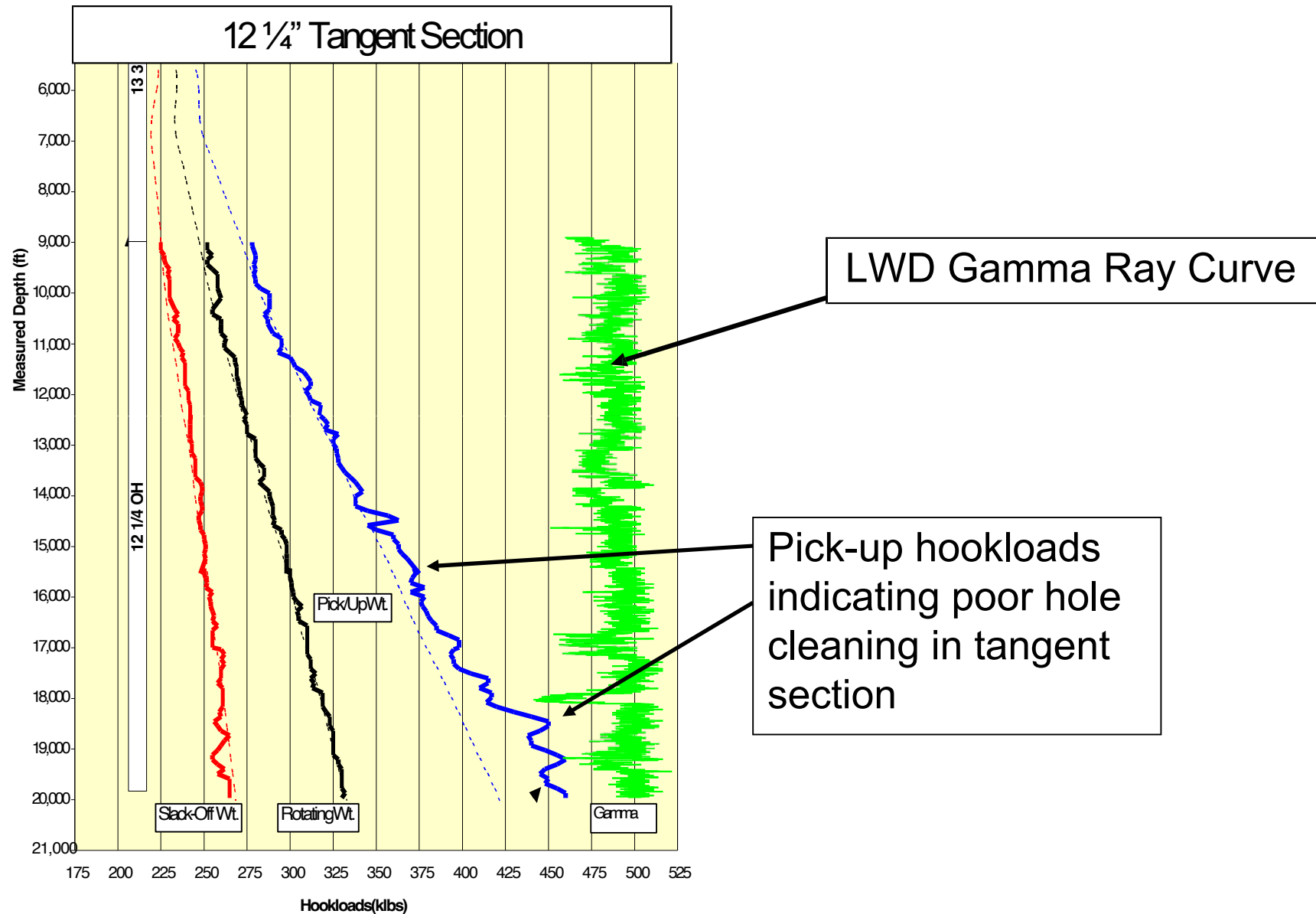
This hookload needs to be used in the T&D charts

Torque & Drag

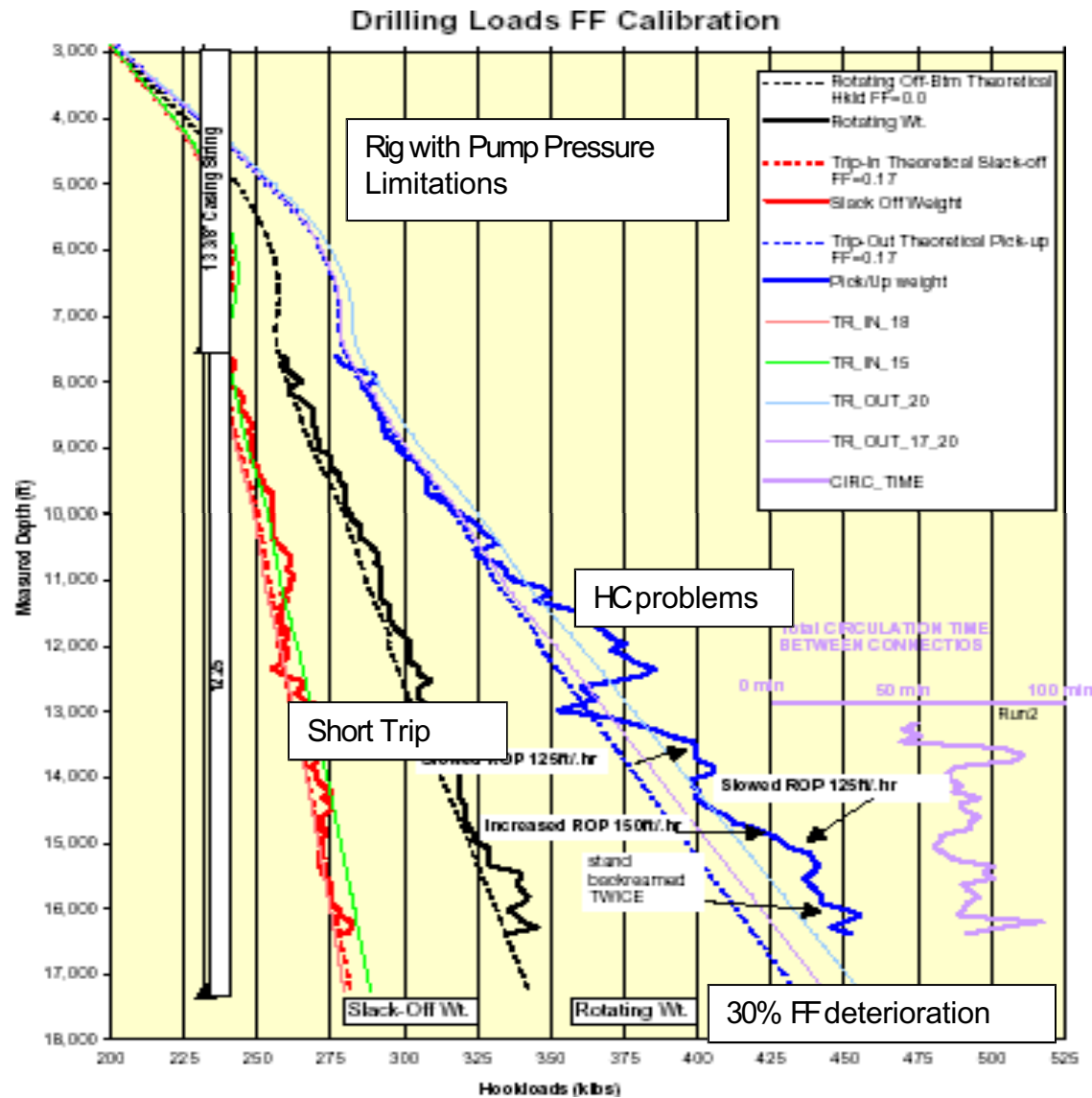
Examples

Hole Condition Monitoring

Poor Hole Cleaning Example



Poor Hole Cleaning- Advanced

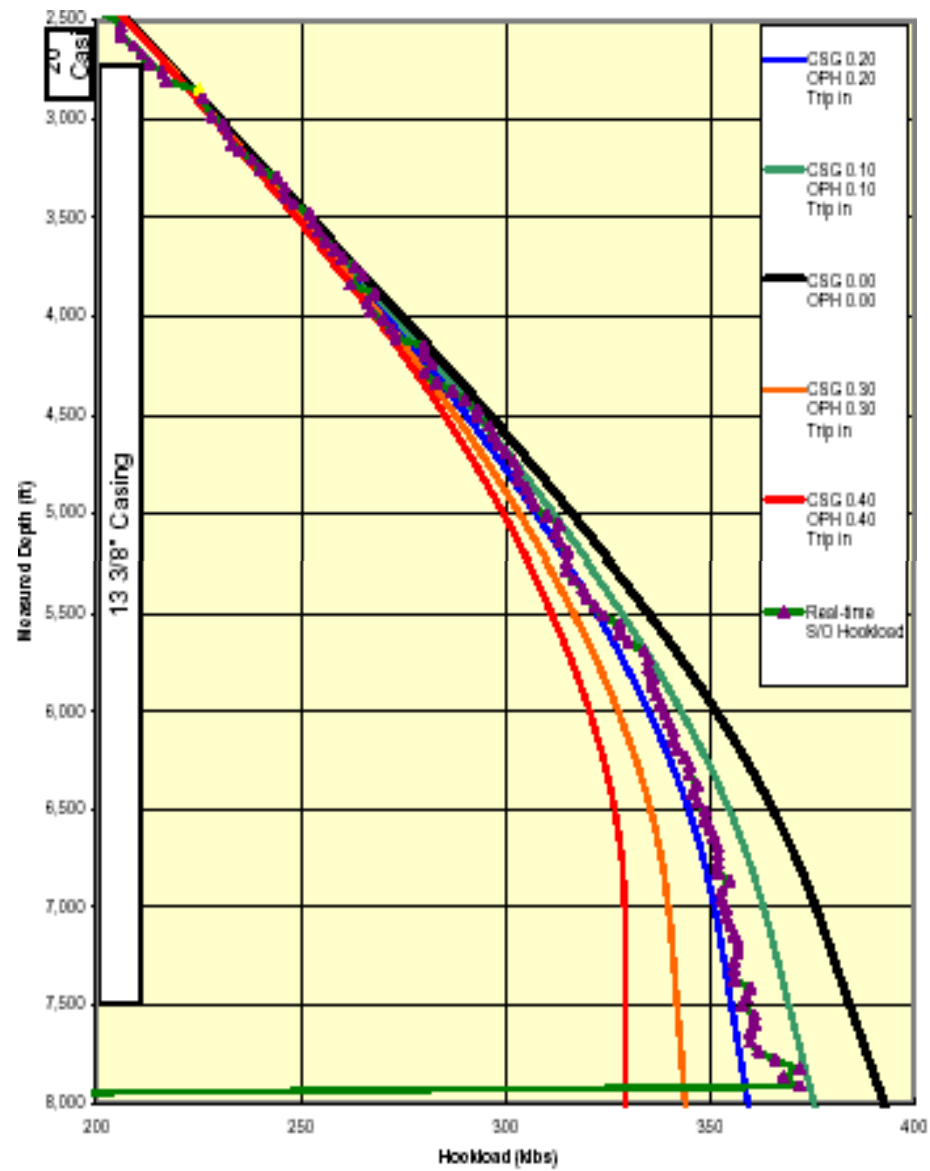


■ 67 degrees

■ Break-outs

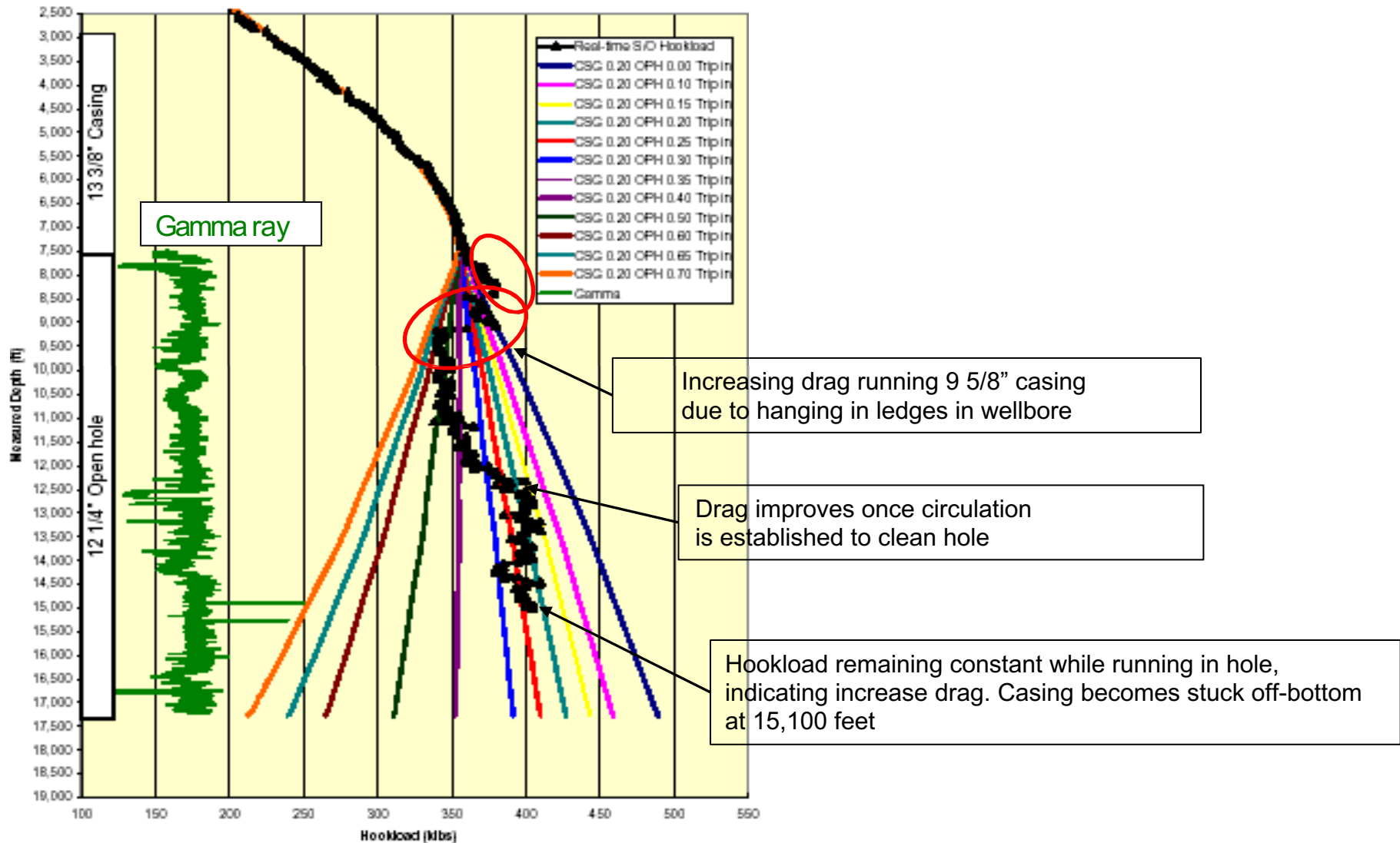


Casing Running - Good



Casing Running - Poor

Csg vs. OH FF Tripping Hookloads



Torque & Drag Management

Further Considerations

Managing Torque and Drag

Torque Reduction

- Well Trajectory
- Cased Hole
- Open Hole
- Mud Lubricity
- Lubricating Beads
- Use of LCM
- Torque reducers

Well path considerations

- Trajectory
- Bottom hole Assemblies
- Optimum Profile

Drag Optimization

- Well Profile
- Mud Lubricity
- Drill pipe protectors
- Buckling Effects
- Weight Distribution
- Hole Cleaning
- Down hole Motors
- Rotation
- Steerable Rotary Systems

General Guidelines for T&D Optimization

- String design can help overcome existing drag
- Place heaviest Drill String Components in the vertical hole section
- Keep tortuosity and doglegs to a minimum (Optimization of well trajectory)
- Use rotary steerable system if feasible
- Use torque reducing subs where side forces are the highest
- Ensure proper hole cleaning.
- Lubricants can be used to effectively reduce Torque and Drag temporarily.
- Run Torque and Drag simulations at several key depths, not just at TD to monitor hole cleaning
- Torque and Drag are caused by lateral forces and friction in the wellbore
- BHAs should be designed to achieve the desired build/turn tendencies with the maximum amount of rotary drilling.
- Bit torque should be monitored

Torque & Drag Reduction



Questions??